chapter 6

Elementary Logic

. . . if it was so, it might be; and if it were so, it would be: but as it isn't, it ain't. That's logic.
—Tweedledee in *Through the Looking-Glass*
by Lewis Carroll (1832–1898)

Why is it important to study logic? Two things that we continually try to accomplish are to understand and to be understood. We attempt to understand an argument given by someone so that we can agree with the conclusion or, possibly, so that we can say that the reasoning does not make sense. We also attempt to express arguments to others without making a mistake. A formal study of logic will help improve these fundamental communication skills.

Why should a student of computer science study logic? A computer scientist needs logical skills to argue whether a problem can be solved on a machine, to transform statements from everyday language to a variety of computer languages, to argue that a program is correct, and to argue that a program is efficient. Computers are constructed from logic devices and are programmed in a logical fashion. Computer scientists must be able to understand and apply new ideas and techniques for programming, many of which require a knowledge of the formal aspects of logic.

In this chapter we'll discuss the formal character of sentences that contain words like "and," "or," and "not" or a phrase like "if A then B."

chapter guide

6.1 (How Do We Reason?) starts our study of logic with a question. We'll discuss some common things that we all do when we reason, and we'll introduce the general idea of a "calculus" as a thing with which to study logic.

6.2 (Propositional Calculus) introduces the basic notions and notations of elementary logic. We'll discuss the properties of tautology, contradiction, and contingency. We'll introduce the idea of equivalence, and we'll use it to find disjunctive and conjunctive normal forms for formulas.

6.3 (Formal Reasoning) introduces basic techniques of formal reasoning. We'll introduce rules that will allow us to write proofs in a formal manner that still reflect the way we do informal reasoning.

6.4 (Formal Axiom Systems) introduces axiom systems. We'll look at a specific set of three axioms and a single rule that are sufficient to prove any required statement of the propositional calculus.

6.1 How Do We Reason?

How do we reason with each other in our daily lives? We probably make arguments, where an *argument* is a finite sequence of statements called *premises* followed by a single statement called the *conclusion*. The conclusion of the argument normally begins with a word or phrase such as one of the following:

Therefore, So, Thus, Hence, It follows that.

For example, here's an argument that might have been made by Descartes before he made the famous statement, "I exist."

If I think, then I exist.

If I do not think, then I think.

Therefore, I exist.

An argument is said to be *valid* if upon assuming that the premises are true it follows that the conclusion is true. So the validity of an argument does not depend on whether the premises are true, but only on whether the conclusion follows from the assumption that the premises are true. We reason by trying to make valid arguments and by trying to decide whether arguments made by others are valid. But how do we tell whether an argument is valid? It seems that each of us has a personal reasoning system that includes some rules of logic that we don't think about and we don't know where they came from.

For example, the most common rule of logic is called *modus ponens* (Latin for "mode that affirms"), and it works like this: Suppose *A* and *B* are two statements and we assume that *A* and "If *A* then *B*" are both true. We can then conclude that *B* is true. For example, consider the following three sentences:

If it is raining, then there are clouds in the sky.

It is raining.

Therefore, there are clouds in the sky.

We use the modus ponens rule without thinking about it. We certainly learned it when we were children, probably by testing a parent. For example, if a child receives a hug from a parent after performing some action, it might dawn on the child that the hug follows after the action. The parent might reinforce the situation by saying, "If you do that again, then you will be rewarded." Parents often make statements such as: "If you touch that stove burner, then you will burn your finger." After touching the burner, the child probably knows a little bit more about modus ponens. A parent might say, "If you do that again, then you are going to be punished." The normal child probably will do it again and notice that punishment follows. Eventually, in the child's mind, the statement "If . . . then . . . punishment" is accepted as a true statement, and the modus ponens rule has taken root.

Most of us are also familiar with the false reasoning exhibited by the *non sequitur* (Latin for "It does not follow"). For example, someone might make several true statements and then conclude that some other statement is true, even though it does not follow from the preceding statements. The hope is that we can recognize this kind of false reasoning so that we never use it. For example, we can probably agree that the following four sentences form a non sequitur:

You squandered the money entrusted to you.
You did not keep required records.
You incurred more debt than your department is worth.
Therefore, you deserve a promotion.

When two people disagree on what they assume to be true or on how they reason about things, then they have problems trying to reason with each other. Some people call this "lack of communication." Other people call it something worse, especially when things like non sequiturs are part of a person's reasoning system. Can common ground be found? Are there any reasoning systems that are, or should be, contained in everyone's personal reasoning system? The answer is yes. The study of logic helps us understand and describe the fundamental parts of all reasoning systems.

What Is a Calculus?

The Romans used small beads called "calculi" to perform counting tasks. The word "calculi" is the plural of the word "calculus." So it makes sense to think that "calculus" has something to do with calculating. Since there are many kinds of calculation, it shouldn't surprise us that "calculus" is used in many different contexts. Let's give a definition.

A *calculus* is a language of expressions of some kind, with definite rules for forming the expressions. There are values, or meanings, associated with the expressions, and there are definite rules to transform one expression into another expression having the same value.

The English language is something like a calculus, where the expressions are sentences formed by English grammar rules. Certainly, we associate meanings with English sentences. But there are no definite rules for transforming one sentence into another. So our definition of a calculus is not quite satisfied. Let's try again with a programming language X. We'll let the expressions be the programs written in the X language. Is this a calculus? Well, there are certainly rules for forming the expressions, and the expressions certainly have meaning. Are there definite rules for transforming one X language program into another X language program? For most modern programming languages the answer is no. So we don't quite have a calculus. We should note that compilers transform X language programs into Y language programs, where X and Y are different languages. Thus a compiler does not qualify as a calculus transformation rule.

In mathematics the word "calculus" usually means the calculus of real functions. For example, the two expressions

$$D_x[f(x)g(x)] \quad \text{and} \quad f(x)D_xg(x) + g(x)D_xf(x)$$

are equivalent in this calculus. The calculus of real functions satisfies our definition of a calculus because there are definite rules for forming the expressions and there are definite rules for transforming expressions into equivalent expressions.

We'll be studying some different kinds of "logical" calculi. In a logical calculus the expressions are defined by rules, the values of the expressions are related to the concepts of true and false, and there are rules for transforming one expression into another. We'll start with a question.

How Can We Tell Whether Something Is a Proof?

When we reason with each other, we normally use informal proof techniques from our personal reasoning systems. This brings up a few questions:

What is an informal proof?

What is necessary to call something a proof?

How can I tell whether an informal proof is correct?

Is there a proof system to learn for each subject of discussion?

Can I live my life without all this?

A formal study of logic will provide us with some answers to these questions. We'll find general methods for reasoning that can be applied informally in many different situations. We'll introduce a precise language for expressing arguments formally, and we'll discuss ways to translate an informal argument into a formal argument. This is especially important in computer science, in which formal solutions (programs) are required for informally stated problems.

6.2 Propositional Calculus

To discuss reasoning, we need to agree on some rules and notation about the truth of sentences. A sentence that is either true or false is called a *proposition*. For example, each of the following lines contains a proposition:

P	Q	$\neg P$	$P \vee Q$	$P \wedge Q$	$P \to Q$
T	T	F	T	T	T
T	F	F	T	F	F
F	T	T	T	F	T
F	F	T	F	F	T

Figure 6.1 Truth tables.

For example, each of the following lines contains a proposition:

Winter begins in June in the Southern Hemisphere.
$2 + 2 = 4$.
If it is raining, then there are clouds in the sky.
I may or may not go to a movie tonight.
All integers are even.
There is a prime number greater than a googol.

For this discussion we'll denote propositions by the letters P, Q, and R, possibly subscripted. Propositions can be combined to form more complicated propositions, just the way we combine sentences, using the words "not," "and," "or," and the phrase "if . . . then . . .". These combining operations are often called *connectives*. We'll denote them by the following symbols and words:

$\neg$ not, negation.
$\wedge$ and, conjunction.
$\vee$ or, disjunction.
$\to$ conditional, implication.

Two common ways to read the expression $P \to Q$ are "if P then Q" and "P implies Q." Other less common readings are "Q if P," "P is a sufficient condition for Q," and "Q is a necessary condition for P." P is called the *antecedent* and Q is called the *consequent* of $P \to Q$.

Now that we have some symbols, we can denote propositions in symbolic form. For example, if P denotes the proposition "It is raining" and Q denotes the proposition "There are clouds in the sky," then $P \to Q$ denotes the proposition "If it is raining, then there are clouds in the sky." Similarly, $\neg P$ denotes the proposition "It is not raining."

The four logical operators are defined to reflect their usage in everyday English. Figure 6.1 is a *truth table* that defines the operators for the possible truth values of their operands.

6.2.1 Well-Formed Formulas and Semantics

Like any programming language or any natural language, whenever we deal with symbols, at least two questions always arise. The first deals with syntax: Is an expression grammatically (or syntactically) correct? The second deals with semantics: What is the meaning of an expression? Let's look at the first question first.

A grammatically correct expression is called a *well-formed formula*, or *wff* for short, which can be pronounced "woof." To decide whether an expression is a wff, we need to precisely define the syntax (or grammar) rules for the formation of wffs in our language. So let's do it.

Syntax

As with any language, we must agree on a set of symbols to use as the alphabet. For our discussion we will use the following sets of symbols:

Truth symbols:	T (or True), F (or False)
Connectives:	$\neg, \to, \wedge, \vee$
Propositional variables:	Uppercase letters like P, Q, and R
Punctuation symbols:	(,)

Next we need to define those expressions (strings) that form the wffs of our language. We do this by giving the following informal inductive definition for the set of propositional wffs.

The Definition of a Wff

A wff is either a truth symbol, or a propositional variable, or the negation of a wff, or the conjunction of two wffs, or the disjunction of two wffs, or the implication of one wff from another, or a wff surrounded by parentheses.

For example, the following expressions are wffs:

$$\text{True, False, } P, \neg Q, P \wedge Q, P \rightarrow Q, (P \vee Q) \wedge R, P \wedge Q \rightarrow R.$$

If we need to justify that some expression is a wff, we can apply the inductive definition. Let's look at an example.

example 6.1 Analyzing a Wff

We'll show that the expression $P \wedge Q \vee R$ is a wff. First, we know that P, Q, and R are wffs because they are propositional variables. Therefore, $Q \vee R$ is a wff because it's a disjunction of two wffs. It follows that $P \wedge Q \vee R$ is a wff because it's a conjunction of two wffs. We could have arrived at the same conclusion by saying that $P \wedge Q$ is a wff and then stating that $P \wedge Q \vee R$ is a wff, since it is the disjunction of two wffs.

end example

Semantics

Can we associate a truth table with each wff? Yes we can, once we agree on a hierarchy of precedence among the connectives. For example, $P \wedge Q \vee R$ is a perfectly good wff. But to find a truth table, we need to agree on which connective to evaluate first. We will define the following hierarchy of evaluation for the connectives of the propositional calculus:

$$\begin{array}{ll} \neg & \text{(highest, do first)} \\ \wedge & \\ \vee & \\ \rightarrow & \text{(lowest, do last)} \end{array}$$

We also agree that the operations $\wedge$, $\vee$, and $\rightarrow$ are left associative. In other words, if the same operation occurs two or more times in succession, without parentheses, then evaluate the operations from left to right. Be sure you can tell the reason for each of the following lines, where each line contains a wff together with a parenthesized wff with the same meaning:

$$\begin{array}{lll} P \vee Q \wedge R & \text{means} & P \vee (Q \wedge R). \\ P \rightarrow Q \rightarrow R & \text{means} & (P \rightarrow Q) \rightarrow R. \\ \neg P \vee Q & \text{means} & (\neg P) \vee Q. \\ \neg P \rightarrow P \wedge Q \vee R & \text{means} & (\neg P) \rightarrow ((P \wedge Q) \vee R). \\ \neg \neg P & \text{means} & \neg(\neg P). \end{array}$$

Any wff has a natural syntax tree that clearly displays the hierarchy of the connectives. For example, the syntax tree for the wff $P \wedge (Q \vee \neg R)$ is given by the diagram in Figure 6.2.

Now we can say that any wff has a unique truth table. For example, suppose we want to find the truth table for the wff

$$\neg P \rightarrow Q \wedge R.$$

From the hierarchy of evaluation we know that this wff has the following parenthesized form:

$$(\neg P) \rightarrow (Q \wedge R).$$

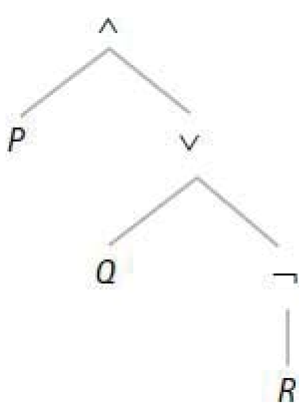

Figure 6.2 Syntax tree.

P	Q	R	$\neg P$	$Q \wedge R$	$\neg P \rightarrow Q \wedge R$
T	T	T	F	T	T
T	T	F	F	F	T
T	F	T	F	F	T
T	F	F	F	F	T
F	T	T	T	T	T
F	T	F	T	F	F
F	F	T	T	F	F
F	F	F	T	F	F

Figure 6.3 Truth table.

So we can construct the truth table as follows: Begin by writing down all possible truth values for the three variables P, Q, and R. This gives us a table with eight lines. Next, compute a column of values for $\neg P$. Then compute a column of values for $Q \wedge R$. Finally, use these two columns to compute the column of values for $\neg P \rightarrow Q \wedge R$. Figure 6.3 gives the result.

Although we've talked some about meaning, we haven't specifically defined the *meaning*, or *semantics*, of a wff. Let's do it now.

The Meaning of a Wff

The meaning of the truth symbol T (or True) is true and the meaning of the truth symbol F (or False) is false. In other words, T (or True) is a truth and F (or False) is a falsity. Otherwise, the meaning of a wff is its truth table.

Tautology, Contradiction, and Contingency

A wff is a *tautology* if its truth table values are all T.

A wff is a *contradiction* if its truth table values are all F.

A wff is a *contingency* if its truth table has at least one T and at least one F.

We have the following fundamental examples:

T (or True) is a tautology.

$P \vee \neg P$ is a tautology.

F (or False) is a contradiction.

$P \wedge \neg P$ is a contradiction.

P is a contingency.

Notational Convenience

We will often use uppercase letters to refer to arbitrary propositional wffs. For example, if we say, "A is a wff," we mean that A represents some arbitrary wff. We also use uppercase letters to denote specific propositional wffs. For example, if we want to talk about the wff $P \wedge (Q \vee \neg R)$ several times in a discussion, we might let $W = P \wedge (Q \vee \neg R)$. Then we can refer to W instead of always writing down the symbols $P \wedge (Q \vee \neg R)$.

6.2.2 Logical Equivalence

In our normal discourse we often try to understand a sentence by rephrasing it in some way. Of course, we always want to make sure that the two sentences have the same meaning. This idea carries over to formal logic too, where we want to describe the idea of equivalence between two wffs.

Definition of Logical Equivalence

Two wffs A and B are *logically equivalent* (or *equivalent*) if they have the same truth value for each assignment of truth values to the set of all propositional variables occurring in the wffs. In this case we write

$$A \equiv B.$$

If two wffs contain the same propositional variables, then they will be equivalent if and only if they have the same truth tables. For example, the wffs

$\neg P \vee Q$ and $P \rightarrow Q$ both contain the propositional variables P and Q. The truth tables for the two wffs are shown in Figure 6.4. Since the tables are the same, we have $\neg P \vee Q \equiv P \rightarrow Q$.

Two wffs that do not share the same propositional variables can still be equivalent. For example, the wffs $\neg P$ and $\neg P \vee (Q \wedge \neg P)$ don't share Q. Since the truth table for $\neg P$ has two lines and the truth table for $\neg P \vee (Q \wedge \neg P)$ has four lines, the two truth tables can't be the same. But we can still compare the truth values of the wffs for each truth assignment to the variables that occur in both wffs. We can do this with a truth table using the variables P and Q as shown in Figure 6.5. Since the columns agree, we know the wffs are equivalent. So we have $\neg P \equiv \neg P \vee (Q \wedge \neg P)$.

P	Q	$\neg P \vee Q$	$P \rightarrow Q$
T	T	T	T
T	F	F	F
F	T	T	T
F	F	T	T

Figure 6.4 Equivalent wffs.

P	Q	$\neg P$	$\neg P \vee (Q \wedge \neg P)$
T	T	F	F
T	F	F	F
F	T	T	T
F	F	T	T

Figure 6.5 Equivalent wffs.

Basic Equivalences (6.1)

Negation	*Disjunction*	*Conjunction*	*Implication*
$\neg\neg A \equiv A$	$A \vee \text{True} \equiv \text{True}$	$A \wedge \text{True} \equiv A$	$A \to \text{True} \equiv \text{True}$
	$A \vee \text{False} \equiv A$	$A \wedge \text{False} \equiv \text{False}$	$A \to \text{False} \equiv \neg A$
	$A \vee A \equiv A$	$A \wedge A \equiv A$	$\text{True} \to A \equiv A$
	$A \vee \neg A \equiv \text{True}$	$A \wedge \neg A \equiv \text{False}$	$\text{False} \to A \equiv \text{True}$
			$A \to A \equiv \text{True}$

Some Conversions	*Absorption laws*
$A \to B \equiv \neg A \vee B$	$A \wedge (A \vee B) \equiv A$
$\neg(A \to B) \equiv A \wedge \neg B$	$A \vee (A \wedge B) \equiv A$
$A \to B \equiv A \wedge \neg B \to \text{False}$	$A \wedge (\neg A \vee B) \equiv A \wedge B$
$\wedge$ and $\vee$ are associative.	$A \vee (\neg A \wedge B) \equiv A \vee B$
$\wedge$ and $\vee$ are commutative.	
$\wedge$ and $\vee$ distribute over each other:	*De Morgan's Laws*
$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$	$\neg(A \wedge B) \equiv \neg A \vee \neg B$
$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$	$\neg(A \vee B) \equiv \neg A \wedge \neg B$

Figure 6.6 Equivalences, conversions, basic laws.

When two wffs don't have any propositional variables in common, the only way for them to be equivalent is that they are either both tautologies or both contradictions. Can you see why? For example, $P \vee \neg P \equiv Q \to Q \equiv \text{True}$.

The definition of equivalence also allows us to make the following useful formulation in terms of conditionals and tautologies.

Equivalence

$A \equiv B$ if and only if $(A \to B) \wedge (B \to A)$ is a tautology

if and only if $A \to B$ and $B \to A$ are tautologies.

Before we go much further, let's list a few basic equivalences. Figure 6.6 shows a collection of equivalences, all of which are easily verified by truth tables, so we'll leave them as exercises.

Reasoning with Equivalences

Can we do anything with the basic equivalences? Sure. We can use them to show that other wffs are equivalent without checking truth tables. But first we need to observe two general properties of equivalence.

The first thing to observe is that equivalence is an "equivalence" relation. In other words, $\equiv$ satisfies the reflexive, symmetric, and transitive properties. The transitive property is the most important property for our purposes. It can be stated as follows for any wffs W, X, and Y:

$$\text{If } W \equiv X \text{ and } X \equiv Y, \text{ then } W \equiv Y.$$

This property allows us to write a sequence of equivalences and then conclude that the first wff is equivalent to the last wff, just the way we do it with ordinary equality of algebraic expressions.

The next thing to observe is the *replacement rule* for equivalences, which is similar to the old rule: "You can always replace equals for equals."

> **Replacement Rule**
>
> If a wff W is changed by replacing a subwff (i.e., a wff that is part of W) by an equivalent wff, then the wff obtained in this way is equivalent to W.

Can you see why this is OK for equivalences? For example, suppose we want to simplify the wff $B \rightarrow (A \vee (A \wedge B))$. We might notice that one of the laws from (6.1) gives $A \vee (A \wedge B) \equiv A$. Therefore, we can apply the replacement rule and write the following equivalence:

$$B \rightarrow (A \vee (A \wedge B)) \equiv B \rightarrow A.$$

Let's do an example to illustrate the process of showing that two wffs are equivalent without checking truth tables.

example 6.2 A Conditional Relationship

The following equivalence shows an interesting relationship involving the connective $\to$.

$$A \to (B \to C) \equiv B \to (A \to C).$$

We'll prove it using equivalences that we already know. Make sure you can give the reason for each line of the proof.

$$\begin{aligned} \text{Proof:} \quad A \to (B \to C) &\equiv A \to (\neg B \vee C) \\ &\equiv \neg A \vee (\neg B \vee C) \\ &\equiv (\neg A \vee \neg B) \vee C \\ &\equiv (\neg B \vee \neg A) \vee C \\ &\equiv \neg B \vee (\neg A \vee C) \\ &\equiv B \to (\neg A \vee C) \\ &\equiv B \to (A \to C). \text{ QED.} \end{aligned}$$

end example

This example illustrates that we can use known equivalences like (6.1) as rules to transform wffs into other wffs that have the same meaning. This justifies the word "calculus" in the name "propositional calculus."

We can also use known equivalences to prove that a wff is a tautology. Here's an example.

example 6.3 An Equivalence with Descartes

We'll prove the validity of the following argument that, as noted earlier, might have been made by Descartes.

If I think, then I exist.
If I do not think, then I think.
Therefore, I exist.

We can formalize the argument by letting A and B be "I think" and "I exist," respectively. Then the argument can be represented by saying that the conclusion B follows from the premises $A \to B$ and $\neg A \to A$. A truth table for the following wff can then be used to show the argument is valid.

$$(A \to B) \wedge (\neg A \to A) \to B.$$

But we'll prove that the wff is a tautology by using basic equivalences to show it is equivalent to True. Make sure you can give the reason for each line of the proof.

Proof:
$$\begin{aligned}(A \to B) \wedge (\neg A \to A) \to B &\equiv (\neg A \vee B) \wedge (\neg\neg A \vee A) \to B \\ &\equiv (\neg A \vee B) \wedge (A \vee A) \to B \\ &\equiv (\neg A \vee B) \wedge A \to B \\ &\equiv B \wedge A \to B \\ &\equiv \neg\ (B \wedge A) \vee B \\ &\equiv \neg B \vee \neg A \vee B \\ &\equiv \text{True} \vee \neg A \\ &\equiv \text{True.} \qquad \text{QED.}\end{aligned}$$

end example

Is It a Tautology, a Contradiction, or a Contingency?

Suppose our task is to find whether a wff W is a tautology, a contradiction, or a contingency. If W contains n variables, then there are 2^n different assignments of truth values to the variables of W. Building a truth table with 2^n rows can be tedious when n is moderately large.

Are there any other ways to determine the meaning of a wff? Yes. One way is to use equivalences to transform the wff into a wff that we recognize as a tautology, a contradiction, or a contingency. But another way, called Quine's method, combines the substitution of variables with the use of equivalences. To describe the method we need a definition.

Definition

If A is a variable in the wff W, then the expression $W(A/\text{True})$ denotes the wff obtained from W by replacing all occurrences of A by True. Similarly, we define $W(A/\text{False})$ to be the wff obtained from W by replacing all occurrences of A by False.

For example, if $W = (A \rightarrow B) \wedge (A \rightarrow C)$, then $W(A/\text{True})$ and $W(A/\text{False})$ have the following values, where we've continued in each case with some basic equivalences.

$$W\,(A/\text{True}) = (\text{True} \rightarrow B) \wedge (\text{True} \rightarrow C) \equiv B \wedge C.$$
$$W\,(A/\text{False}) = (\text{False} \rightarrow B) \wedge (\text{False} \rightarrow C) \equiv \text{True} \wedge \text{True} \equiv \text{True}.$$

Now comes the key observation that allows us to use these ideas to decide the truth value of a wff.

Substitution Properties

1. W is a tautology iff $W(A/\text{True})$ and $W(A/\text{False})$ are tautologies.
2. W is a contradiction iff $W(A/\text{True})$ and $W(A/\text{False})$ are contradictions.

For example, in our little example we found that $W(A/\text{True}) \equiv B \wedge C$, which is a contingency, and $W(A/\text{False}) \equiv \text{True}$, which is a tautology. Therefore, W is a contingency.

The idea of Quine's method is to construct $W(A/\text{True})$ and $W(A/\text{False})$ and then to simplify these wffs by using the basic equivalences. If we can't tell the truth values, then choose another variable and apply the method to each of these wffs. A complete example is in order.

example 6.4 Quine's Method

Suppose that we want to check the meaning of the following wff W:

$$[(A \wedge B \rightarrow C) \wedge (A \rightarrow B)] \rightarrow (A \rightarrow C).$$

First we compute the two wffs $W(A/\text{True})$ and $W(A/\text{False})$ and simplify them using basic equivalences where appropriate.

$$\begin{aligned} W(A/\text{True}) &= [(\text{True} \wedge B \to C) \wedge (\text{True} \to B)] \to (\text{True} \to C) \\ &\equiv [(B \to C) \wedge (\text{True} \to B)] \to (\text{True} \to C) \\ &\equiv [(B \to C) \wedge B] \to C. \end{aligned}$$

$$\begin{aligned} W(A/\text{False}) &= [(\text{False} \wedge B \to C) \wedge (\text{False} \to B)] \to (\text{False} \to C) \\ &\equiv [(\text{False} \to C) \wedge \text{True}] \to \text{True} \\ &\equiv \text{True}. \end{aligned}$$

Therefore, $W(A/\text{False})$ is a tautology. Now we need to check the simplification of $W(A/\text{True})$. Call it X. We continue the process by constructing the two wffs $X(B/\text{True})$ and $X(B/\text{False})$:

$$\begin{aligned} X(B/\text{True}) &= [(\text{True} \to C) \wedge \text{True}] \to C \\ &\equiv [C \wedge \text{True}] \to C \\ &\equiv C \to C \\ &\equiv \text{True}. \end{aligned}$$

So $X(B/\text{True})$ is a tautology. Now let's look at $X(B/\text{False})$.

$$\begin{aligned} X(B/\text{False}) &= [(\text{False} \to C) \wedge \text{False}] \to C \\ &\equiv [\text{True} \wedge \text{False}] \to C \\ &\equiv \text{False} \to C \\ &\equiv \text{True}. \end{aligned}$$

So $X(B/\text{False})$ is also a tautology. Therefore, X is a tautology and it follows that W is a tautology.

end example

Quine's method can also be described graphically with a binary tree. Let W be the root. If N is any node, pick one of its variables, say V, and let the two children of N be $N(V/\text{True})$ and $N(V/\text{False})$. Each node should be simplified as much as possible. Then W is a tautology if all leaves are True, a contradiction if all leaves are False, and a contingency otherwise. Let's illustrate the idea with the wff $P \to Q \wedge P$. The binary tree in Figure 6.7 shows that the wff $P \to Q \wedge P$ is a contingency because Quine's method gives one False leaf and two True leaves.

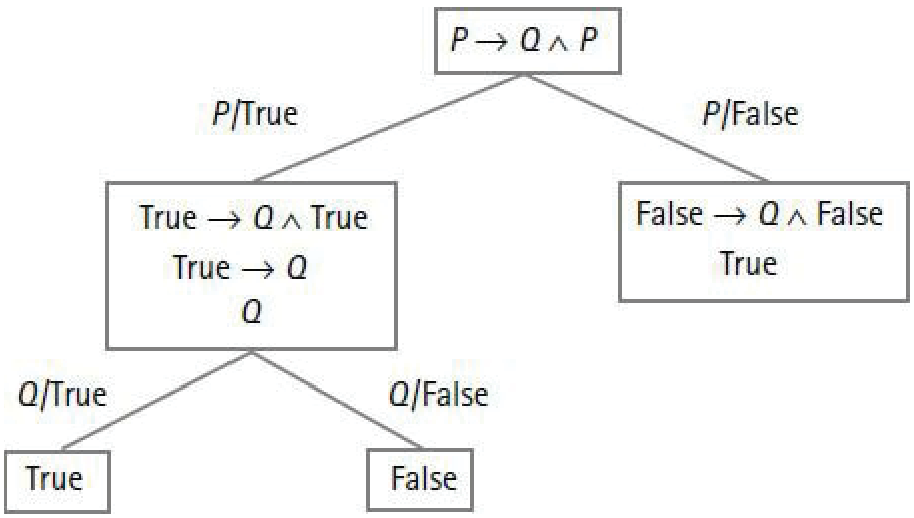

Figure 6.7 Quine's method.

6.2.3 Truth Functions and Normal Forms

A *truth function* is a function with a finite number of arguments, where the argument values and function values are from the set {True, False}. So any wff defines a truth function. For example, the function g defined by

$$g(P, Q) = P \wedge Q$$

is a truth function. Is the converse true? In other words, is every truth function a wff? The answer is yes. To see why this is true, we'll present a technique to construct a wff for any truth function.

For example, suppose we define a truth function f by saying that $f(P, Q)$ is true exactly when P and Q have opposite truth values. Is there a wff that has the same truth table as f? We'll introduce the technique with this example. Figure 6.8 is the truth table for f. We'll explain the statements on the right side of this table.

We've written the two wffs $P \wedge \neg Q$ and $\neg P \wedge Q$ on the second and third lines of the table because f has the value T on these lines. Each wff is a conjunction of argument variables or their negations according to their values on the same line subject to the following two rules:

If P has the value T, then put P in the conjunction.

If P has the value F, then put $\neg P$ in the conjunction.

Let's see why we want to follow these rules. Notice that the truth table for $P \wedge \neg Q$ in Figure 6.9 has exactly one T and it occurs on the second line.

P	Q	$f(P, Q)$	*Action*
T	T	F	
T	F	T	Create $P \wedge \neg Q$
F	T	T	Create $\neg P \wedge Q$
F	F	F	

Figure 6.8 A truth function.

P	Q	$f(P, Q)$	$P \wedge \neg Q$	$\neg P \wedge Q$
T	T	F	F	F
T	F	T	T	F
F	T	T	F	T
F	F	F	F	F

Figure 6.9 A truth function.

Similarly, the truth table for $\neg P \wedge Q$ has exactly one T and it occurs on the third line of the table.

So each of the tables for $P \wedge \neg Q$ and $\neg P \wedge Q$ has exactly one T per column, and they occur on the same lines where f has the value T. Since there is one conjunctive wff for each occurrence of T in the table for f, it follows that the table for f can be obtained by taking the disjunction of the tables for $P \wedge \neg Q$ and $\neg P \wedge Q$. Thus we obtain the following equivalence.

$$f(P, Q) \equiv (P \wedge \neg Q) \vee (\neg P \wedge Q).$$

Let's do another example to get the idea. Then we'll discuss the special forms that we obtain by using this technique.

example 6.5 Converting a Truth Function

Let f be the truth function defined as follows:

$$f(P, Q, R) = \text{True if and only if either } P = Q = \text{False or } Q = R = \text{True}.$$

Then f is true in exactly the following four cases:

f(F, F, T),
f(F, F, F),
f(T, T, T),
f(F, T, T).

So we can construct a wff equivalent to f by taking the disjunction of the four wffs that correspond to these four cases. The disjunction follows.

$$(\neg P \wedge \neg Q \wedge R) \vee (\neg P \wedge \neg Q \wedge \neg R) \vee (P \wedge Q \wedge R) \vee (\neg P \wedge Q \wedge R).$$

end example

The method we have described can be generalized to construct an equivalent wff for any truth function with True for at least one if its values. If a truth function doesn't have any True values, then it is a contradiction and is equivalent to False. So every truth function is equivalent to some propositional wff. We'll state this as the following theorem:

Truth Functions **(6.2)**

Every truth function is equivalent to a propositional wff defined in terms of the connectives $\neg$, $\wedge$, and $\vee$.

Now we're going to discuss some useful forms for propositional wffs. But first we need a little terminology. A *literal* is a propositional variable or its negation. For example, P, Q, $\neg P$, and $\neg Q$ are literals.

Disjunctive Normal Form

A *fundamental conjunction* is either a literal or a conjunction of two or more literals. For example, P and $P \wedge \neg Q$ are fundamental conjunctions. A *disjunctive normal form* (DNF) is either a fundamental conjunction or a disjunction of two or more fundamental conjunctions. For example, the following wffs are DNFs:

$$P \vee (\neg P \wedge Q),$$
$$(P \wedge Q) \vee (\neg Q \wedge P),$$
$$(P \wedge Q \wedge R) \vee (\neg P \wedge Q \wedge R).$$

Sometimes the trivial cases are hardest to see. For example, try to explain why the following four wffs are DNFs: P, $\neg P$, $P \vee \neg P$, and $\neg P \wedge Q$. The propositions that we constructed for truth functions are DNFs.

It is often the case that a DNF is equivalent to a simpler DNF. For example, the DNF $P \vee (P \wedge Q)$ is equivalent to the simpler DNF P by using (6.1). For another example, consider the following DNF:

$$(P \wedge Q \wedge R) \vee (\neg P \wedge Q \wedge R) \vee (P \wedge R).$$

The first fundamental conjunction is equivalent to $(P \wedge R) \wedge Q$, which we see contains the third fundamental conjunction $P \wedge R$ as a subexpression. Thus the first term of the DNF can be absorbed by (6.1) into the third term, which gives the following simpler equivalent DNF:

$$(\neg P \wedge Q \wedge R) \vee (P \wedge R).$$

For any wff W we can always construct an equivalent DNF. If W is a contradiction, then it is equivalent to the single term DNF $P \wedge \neg P$. If W is not a contradiction, then we can write down its truth table and use the technique that we used for truth functions to construct a DNF. So we can make the following statement.

Disjunctive Normal Form **(6.3)**

Every wff is equivalent to a DNF.

Another way to construct a DNF for a wff is to transform it into a DNF by using the equivalences of (6.1). In fact we'll outline a short method that will always do the job:

First, remove all occurrences (if there are any) of the connective $\rightarrow$ by using the equivalence

$$A \rightarrow B \equiv \neg A \vee B.$$

Next, move all negations inside to create literals by using De Morgan's equivalences

$$\neg (A \wedge B) \equiv \neg A \vee \neg B \text{ and } \neg (A \vee B) \equiv \neg A \wedge \neg B.$$

Finally, apply the distributive equivalences to obtain a DNF. Let's look at an example.

example 6.6 A DNF Construction

We'll construct a DNF for the wff $((P \wedge Q) \rightarrow R) \wedge S$.

$$\begin{aligned}((P \wedge Q) \rightarrow R) \wedge S &\equiv (\neg (P \wedge Q) \vee R) \wedge S \\ &\equiv (\neg P \vee \neg Q \vee R) \wedge S \\ &\equiv (\neg P \wedge S) \vee (\neg Q \wedge S) \vee (R \wedge S).\end{aligned}$$

end example

Suppose W is a wff having n distinct propositional variables. A DNF for W is called a *full disjunctive normal form* if each fundamental conjunction has exactly n literals, one for each of the n variables appearing in W. For example, the following wff is a full DNF:

$$(P \wedge Q \wedge R) \vee (\neg P \wedge Q \wedge R).$$

The wff $P \vee (\neg P \wedge Q)$ is a DNF but not a full DNF because the variable Q does not occur in the first fundamental conjunction.

The truth table technique to construct a DNF for a truth function automatically builds a full DNF because all of the variables in a wff occur in each fundamental conjunction. So we can state the following result.

Full Disjunctive Normal Form **(6.4)**

Every wff that is not a contradiction is equivalent to a full DNF.

Conjunctive Normal Form

In a manner entirely analogous to the previous discussion we can define a *fundamental disjunction* to be either a literal or the disjunction of two or more literals. A *conjunctive normal form* (CNF) is either a fundamental disjunction or a conjunction of two or more fundamental disjunctions. For example, the following wffs are CNFs:

$$P \wedge (\neg P \vee Q),$$
$$(P \vee Q) \wedge (\neg Q \vee P),$$
$$(P \vee Q \vee R) \wedge (\neg P \vee Q \vee R).$$

Let's look at some trivial examples. Notice that the following four wffs are CNFs: P, $\neg P$, $P \wedge \neg P$, and $\neg P \vee Q$. As in the case for DNFs, some CNFs are equivalent to simpler CNFs. For example, the CNF $P \neg (P \vee Q)$ is equivalent to the simpler CNF P by (6.1).

Suppose some wff W has n distinct propositional letters. A CNF for W is called a *full conjunctive normal form* if each fundamental disjunction has exactly n literals, one for each of the n variables that appear in W. For example, the following wff is a full CNF:

$$(P \vee Q \vee R) \wedge (\neg P \vee Q \vee R).$$

On the other hand, the wff $P \wedge (\neg P \vee Q)$ is a CNF but not a full CNF.

It's possible to write any truth function f that is not a tautology as a full CNF. In this case we associate a fundamental disjunction with each line of the truth table in which f has the value False. Let's return to our original example, in which $f(P, Q)$ = True exactly when P and Q have opposite truth values. Figure 6.10 shows the values for f together with a fundamental disjunction created for each line where f has the value False.

In this case, $\neg P$ is added to the disjunction if P = True, and P is added to the disjunction if P = False. Then we take the conjunction of these disjunctions to obtain the following conjunctive normal form of f:

$$f(P, Q) \equiv (\neg P \vee \neg Q) \wedge (P \vee Q).$$

P	Q	$f(P, Q)$	*Action*
T	T	F	Create $\neg P \vee \neg Q$
T	F	T	
F	T	T	
F	F	F	Create $P \vee Q$

Figure 6.10 A truth function.

Of course, any tautology is equivalent to the single term CNF $P \vee \neg P$. Now we can state the following results for CNFs, which correspond to statements (6.3) and (6.4) for DNFs:

Conjunctive Normal Form

Every wff is equivalent to a CNF. (6.5)

Every wff that is not a tautology is equivalent to a full CNF.	(6.6)

We should note that some authors use the terms "disjunctive normal form" and "conjunctive normal form" to describe the expressions that we have called "full disjunctive normal forms" and "full conjunctive normal forms." For example, they do not consider $P \vee (\neg P \wedge Q)$ to be a DNF. We use the more general definitions of DNF and CNF because they are useful in describing methods for automatic reasoning and they are useful in describing methods for simplifying digital logic circuits.

Constructing Full Normal Forms Using Equivalences

We can construct full normal forms for wffs without resorting to truth table techniques. Let's start with the full disjunctive normal form. To find a full DNF for a wff, we first convert it to a DNF by the usual actions: eliminate conditionals, move negations inside, and distribute $\wedge$ over $\vee$. For example, the wff $P \wedge (Q \to R)$ can be converted to a DNF in two steps, as follows:

$$\begin{aligned} P \wedge (Q \to R) &\equiv P \wedge (\neg Q \vee R) \\ &\equiv (P \wedge \neg Q) \vee (P \wedge R). \end{aligned}$$

The right side of the equivalence is a DNF. However, it's not a full DNF because the two fundamental conjunctions don't contain all three variables. The trick to add the extra variables can be described as follows:

Adding a Variable to a Fundamental Conjunction

To add a variable, say R, to a fundamental conjunction C without changing the value of C, write the following equivalences:

$$C \equiv C \wedge \text{True} \equiv C \wedge (R \vee \neg R) \equiv (C \wedge R) \vee (C \wedge \neg R).$$

Let's continue with our example. First, we'll add the letter R to the fundamental conjunction $P \wedge \neg Q$. Be sure to justify each step of the following calculation:

$$\begin{aligned} P \wedge \neg Q &\equiv (P \wedge \neg Q) \wedge \text{True} \\ &\equiv (P \wedge \neg Q) \wedge (R \vee \neg R) \\ &\equiv (P \wedge \neg Q \wedge R) \vee (P \wedge \neg Q \wedge \neg R)\,. \end{aligned}$$

Next, we'll add the variable Q to the fundamental conjunction $P \wedge R$:

$$\begin{aligned} P \wedge R &\equiv (P \wedge R) \wedge \text{True} \\ &\equiv (P \wedge R) \wedge (Q \vee \neg Q) \\ &\equiv (P \wedge R \wedge Q) \vee (P \wedge R \wedge \neg Q)\,. \end{aligned}$$

Lastly, we put the two wffs together to obtain a full DNF for $P \wedge (Q \rightarrow R)$:

$$(P \wedge \neg Q \wedge R) \vee (P \wedge \neg Q \wedge \neg R) \vee (P \wedge R \wedge Q) \vee (P \wedge R \wedge \neg Q).$$

Notice that the wff can be simplified to a full DNF with three fundamental conjunctions.

example 6.7 Constructing a Full DNF

We'll construct a full DNF for the wff $P \rightarrow Q$. Make sure to justify each line of the following calculation.

$$
\begin{aligned}
P \to Q &\equiv \neg P \vee Q \\
&\equiv (\neg P \wedge \text{True}) \vee Q \\
&\equiv (\neg P \wedge (Q \vee \neg Q)) \vee Q \\
&\equiv (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee Q \\
&\equiv (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge \text{True}) \\
&\equiv (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge (P \vee \neg P)) \\
&\equiv (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P) \vee (Q \wedge \neg P) \\
&\equiv (\neg P \wedge Q) \vee (\neg P \wedge \neg Q) \vee (Q \wedge P).
\end{aligned}
$$

end example

We can proceed in an entirely analogous manner to find a full CNF for a wff. The trick in this case is to add variables to a fundamental disjunction without changing its truth value. It goes as follows:

Adding a Variable to a Fundamental Disjunction

To add a variable, say R, to a fundamental disjunction D without changing the value of D, write the following equivalences:

$$D \equiv D \vee \text{False} \equiv D \vee (R \wedge \neg R) \equiv (D \vee R) \wedge (D \vee \neg R).$$

For example, let's find a full CNF for the wff $P \wedge (P \to Q)$. To start off, we put the wff in conjunctive normal form as follows:

$$P \wedge (P \to Q) \equiv P \wedge (\neg P \vee Q).$$

The right side is not a full CNF because the variable Q does not occur in the fundamental disjunction P. So we'll apply the trick to add the variable Q. Make sure you can justify each step in the following calculation:

$$
\begin{aligned}
P \wedge (P \to Q) &\equiv P \wedge (\neg P \vee Q) \\
&\equiv (P \vee \text{False}) \wedge (\neg P \vee Q) \\
&\equiv (P \vee (Q \wedge \neg Q)) \wedge (\neg P \vee Q) \\
&\equiv (P \vee Q) \wedge (P \vee \neg Q) \wedge (\neg P \vee Q).
\end{aligned}
$$

The result is a full CNF that is equivalent to the original wff. Let's do another example.

example 6.8 Constructing a Full CNF

We'll construct a full CNF for $(P \to (Q \vee R)) \wedge (P \vee Q)$. After converting the wff to conjunctive normal form, all we need to do is add the variable R to the fundamental disjunction $P \vee Q$. Here's the transformation:

$$
\begin{aligned}
(P \to (Q \vee R)) \wedge (P \vee Q) &\equiv (\neg P \vee Q \vee R) \wedge (P \vee Q) \\
&\equiv (\neg P \vee Q \vee R) \wedge ((P \vee Q) \vee \text{False}) \\
&\equiv (\neg P \vee Q \vee R) \wedge ((P \vee Q) \vee (R \wedge \neg R)) \\
&\equiv (\neg P \vee Q \vee R) \wedge ((P \vee Q \vee R) \wedge (P \vee Q \vee \neg R)) \\
&\equiv (\neg P \vee Q \vee R) \wedge (P \vee Q \vee R) \wedge (P \vee Q \vee \neg R).
\end{aligned}
$$

end example

6.2.4 Adequate Sets of Connectives

A set of connectives is *adequate* (also *functionally complete*) if every truth function is equivalent to a wff defined in terms of the connectives. We've already seen in (6.2) that every truth function is equivalent to a propositional wff defined in terms of the connectives $\neg$, $\wedge$, and $\vee$. Therefore, $\{\neg, \wedge, \vee\}$ is an adequate set of connectives.

Are there any 2-element adequate sets of connectives? The answer is yes. Consider the two connectives $\neg$ and $\vee$. Since $\{\neg, \wedge, \vee\}$ is adequate, we'll have an adequate set if we can show that $\wedge$ can be written in terms of $\neg$ and $\vee$. This can be seen by the equivalence

$$A \wedge B \equiv \neg (\neg A \vee \neg B).$$

So $\{\neg, \vee\}$ is an adequate set of connectives. Other adequate sets of connectives are $\{\neg, \wedge\}$ and $\{\neg, \rightarrow\}$. We'll leave these as exercises.

Are there any single connectives that are adequate? The answer is yes, but we won't find one among the four basic connectives. There is a connective called the NAND operator, which is short for the "Negation of AND." We'll write NAND in functional form NAND(P, Q), since there is no well-established symbol for it. Figure 6.11 shows the truth table for NAND.

P	Q	NAND (P, Q)
T	T	F
T	F	T
F	T	T
F	F	T

Figure 6.11 A truth table.

P	Q	NOR (P, Q)
T	T	F
T	F	F
F	T	F
F	F	T

Figure 6.12 A truth table.

To see that NAND is adequate, we have to show that the connectives in some adequate set can be defined in terms of it. For example, we can write negation in terms of NAND as follows:

$$\neg P \equiv \text{NAND}(P, P).$$

We'll leave it as an exercise to show that the other connectives can be written in terms of NAND.

Another single adequate connective is the NOR operator. NOR is short for the "Negation of OR." Figure 6.12 shows the truth table for NOR. We'll leave it as an exercise to show that NOR is an adequate connective. NAND and NOR are important because they represent the behavior of two important building blocks for logic circuits.

Exercises

Syntax and Semantics

1. Write down the parenthesized version of each of the following expressions.

a. $\neg P \wedge Q \rightarrow P \vee R.$
b. $P \vee \neg Q \wedge R \rightarrow P \vee R \rightarrow \neg Q.$
c. $A \rightarrow B \vee \neg C \wedge D \wedge E \rightarrow F.$

2. Remove as many parentheses as possible from each of the following wffs.

a. $(((P \vee Q) \rightarrow (\neg R)) \vee (((\neg Q) \wedge R) \wedge P)).$
b. $((A \rightarrow (B \vee C)) \rightarrow (A \vee (\neg (\neg B)))).$

3. Let A, B, and C be propositional wffs. Find two different wffs, where the statement "If A then B else C" reflects the meaning of each wff.

Equivalence

4. Use truth tables to verify the equivalences in (6.1).

5. Use other equivalences to prove the equivalence

$$A \rightarrow B \equiv A \wedge \neg B \rightarrow \text{False}.$$

Hint: Start with the right side.

6. Show that $\rightarrow$ is not associative. That is, show that $(A \rightarrow B) \rightarrow C$ is not equivalent to $A \rightarrow (B \rightarrow C)$.

7. Use Quine's method to show that each wff is a contingency.

a. $A \vee B \to B$.

b. $(A \to B) \wedge (B \to \neg A) \to A$.

c. $(A \to B) \wedge (B \to C) \to (C \to A)$.

d. $(A \vee B \to C) \wedge A \to (C \to B)$.

e. $(A \to B) \vee ((C \to \neg B) \wedge \neg C)$.

f. $(A \vee B) \to (C \vee A) \wedge (\neg C \vee B)$.

8. Use Quine's method to show that each wff is a tautology.

a. $(A \to B) \wedge (B \to C) \to (A \to C)$.

b. $(A \vee B) \wedge (A \to C) \wedge (B \to D) \to (C \vee D)$.

c. $(A \to C) \wedge (B \to D) \wedge (\neg C \vee \neg D) \to (\neg A \vee \neg B)$.

d. $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$.

e. $(\neg B \to \neg A) \to ((\neg B \to A) \to B)$.

f. $(A \to B) \to (C \vee A \to C \vee B)$.

g. $(A \to C) \to ((B \to C) \to (A \vee B \to C))$.

h. $(A \to B) \to (\neg (B \wedge C) \to \neg (C \wedge A))$.

9. Verify each of the following equivalences by writing an equivalence proof. That is, start on one side and use known equivalences to get to the other side.

a. $(A \to B) \wedge (A \vee B) \equiv B$.

b. $A \wedge B \to C \equiv (A \to C) \vee (B \to C)$.

c. $A \wedge B \to C \equiv A \to (B \to C)$.

d. $A \vee B \to C \equiv (A \to C) \wedge (B \to C)$.

e. $A \to B \wedge C \equiv (A \to B) \wedge (A \to C)$.

f. $A \to B \vee C \equiv (A \to B) \vee (A \to C)$.

10. Show that each wff is a tautology by using equivalences to show that each wff is equivalent to True.

a. $A \to A \vee B$.

b. $A \wedge B \to A$.

c. $(A \vee B) \wedge \neg A \to B$.

d. $A \to (B \to A)$.
e. $(A \to B) \wedge \neg B \to \neg A$.
f. $(A \to B) \wedge A \to B$.
g. $A \to (B \to (A \wedge B))$.
h. $(A \to B) \to ((A \to \neg B) \to \neg A)$.

Normal Forms

11. Use equivalences to transform each of the following wffs into a DNF.
a. $(P \to Q) \to P$.
b. $P \to (Q \to P)$.
c. $Q \wedge \neg P \to P$.
d. $(P \vee Q) \wedge R$.
e. $P \to Q \wedge R$.
f. $(A \vee B) \wedge (C \to D)$.

12. Use equivalences to transform each of the following wffs into a CNF.
a. $(P \to Q) \to P$.
b. $P \to (Q \to P)$.
c. $Q \wedge \neg P \to P$.
d. $(P \vee Q) \wedge R$.
e. $P \to Q \wedge R$.
f. $(A \wedge B) \vee E \vee F$.
g. $(A \wedge B) \vee (C \wedge D) \vee (E \to F)$.

13. For each of the following functions, write down the full DNF and full CNF representations.
a. $f(P, Q)$ = True if and only if P is True.
b. $f(P, Q, R)$ = True if and only if either Q is True or R is False.

14. Transform each of the following wffs into a full DNF if possible.
a. $(P \to Q) \to P$.
b. $Q \wedge \neg P \to P$.

c. $P \to (Q \to P)$.
d. $(P \vee Q) \wedge R$.
e. $P \to Q \wedge R$.

15. Transform each of the following wffs into a full CNF if possible.

a. $(P \to Q) \to P$.
b. $P \to (Q \to P)$.
c. $Q \wedge \neg P \to P$.
d. $P \to Q \wedge R$.
e. $(P \vee Q) \wedge R$.

Challenges

16. Show that each of the following sets of operations is an adequate set of connectives.

a. $\{\neg, \wedge\}$.
b. $\{\neg, \to\}$.
c. {false, $\to$}.
d. {NAND}.
e. {NOR}.

17. Show that there are no adequate single binary connectives other than NAND and NOR. *Hint:* Let f be the truth function for an adequate binary connective. Show that f(True, True) = False and f(False, False) = True because the negation operation must be represented in terms of f. Then consider the remaining cases in the truth table for f.

6.3 Formal Reasoning

We have seen that truth tables are suffcient to find the truth of any proposition. However, if a proposition contains several connectives, then a truth table can become quite complicated. When we use an equivalence proof rather than truth tables to decide the equivalence of two wffs, it seems a bit closer to the way we communicate with each other. But the usual way we reason is a bit more informal.

We reason informally by writing sentences in English mixed with symbols and expressions from some domain of discourse and try to make conclusions based on assumptions. Now we're going to focus on the formal structure of proof, where the domain of discourse consists of propositions. Although there is no need to formally reason about the truth of propositions, many parts of logic need tools other than truth tables to determine the truth of wffs. Formal reasoning tools for propositional calculus also carry over to other areas of discourse that use logic. So let's get started.

An *argument* is a finite sequence of wffs called *premises* followed by a single wff called the *conclusion*. An argument is *valid* if upon assuming that the premises are true it follows that the conclusion is true. For example, the following argument is valid.

Premises: $A \vee B$,

$\neg A$,

$B \to C$,

Conclusion: $B \wedge C$.

This argument is valid because if the premises $A \vee B$, $\neg A$, and $B \to C$ are true, then the conclusion $B \wedge C$ is true, which can be verified by a truth table.

Our goal is to show that an argument is valid by using a formal proof process that uses a specific set of rules. There are various ways to choose the rules depending on how close we want the formal proof process to approximate the way we reason informally. We'll introduce two approaches to formal proof. In this section we'll study the *natural deduction* approach, where the rules reflect the informal (i.e., natural) way that we reason. In Section 6.4 we'll introduce the *axiomatic* approach, where the premises for arguments are limited to a fixed set of wffs, called *axioms*, and where there is only one rule, modus ponens. When we apply logic to a particular subject, an axiom might also be a statement about the subject under study. For example, "On any two distinct points there is always a line," is an axiom for Euclidean geometry.

6.3.1 Proof Rules

A *proof* (or *derivation*) is a finite sequence of wffs, where each wff is either a premise or the result of applying a rule to certain previous wffs in the sequence. The rules are often called *proof rules* or *inference* rules.

A proof rule gives a condition for which a new wff can be added to a proof. The condition can be a listing of one or more specific wffs or it can be a derivation from a specific premise to a specific conclusion. We'll represent a proof rule by an expression of the form

$$\frac{\text{condition}}{\text{wff}}.$$

We say that the condition of the rule *infers* the wff.

For example, let's look at the modus ponens rule. If we find A and $A \to B$ in a proof and we apply the rule, then B gets added to the proof. Since the order of occurrence of A and $A \to B$ does not matter, we can represent the modus ponens rule by either

$$\frac{A, A \to B}{B} \text{ or } \frac{A \to B, A}{B}.$$

Before we go on, let's list the proof rules that we'll be using to construct proofs. We'll discuss their properties as we go along.

The Proof Rules **(6.7)**

<table>
<tr><td colspan="3">Conjunction (Conj)
$\dfrac{A, B}{A \wedge B}.$</td><td colspan="3">Simplification (Simp)
$\dfrac{A \wedge B}{A}$ and $\dfrac{A \wedge B}{B}.$</td></tr>
<tr><td colspan="3">Addition (Add)
$\dfrac{A}{A \vee B}$ and $\dfrac{A}{B \vee A}.$</td><td colspan="3">Disjunctive Syllogism (DS)
$\dfrac{A \vee B, \neg A}{B}$ and $\dfrac{A \vee B, \neg B}{A}.$</td></tr>
<tr><td colspan="3">Modus Ponens (MP)
$\dfrac{A, A \to B}{B}.$</td><td colspan="3">Conditional Proof (CP)
$\dfrac{\text{From } A, \text{ derive } B}{A \to B}$</td></tr>
<tr><td colspan="2">Double Negation (DN)
$\dfrac{\neg \neg A}{A}$ and $\dfrac{A}{\neg \neg A}.$</td><td colspan="2">Contradiction (Contr)
$\dfrac{A, \neg A}{\text{False}}.$</td><td colspan="2">Indirect Proof (IP)
$\dfrac{\text{From } \neg A, \text{ derive False}}{A}.$</td></tr>
</table>

The seven rules Conj, Simp, Add, DS, MP, DN, and Contr are all valid arguments that come from simple tautologies. For example, the modus ponens rule, MP, infers B from A and $A \to B$. It's easy to check (say, with a truth table) that $A \wedge (A \to B) \to B$ is a tautology. It follows that the truth of the premises A and $A \to B$ implies the truth of the conclusion B. Therefore, MP is a valid argument. The same reasoning applies to the other six rules.

The condition for the CP rule is "From A, derive B," which is short for "There is a derivation from the premise A that ends with B." If we find such a derivation in a proof and apply the CP rule, then $A \to B$ is added to the proof. Similarly, the condition for the IP rule is "From $\neg W$, derive False," which is short for "There is a derivation from the premise $\neg W$ that ends with a contradiction." If we find such a derivation in a proof and apply the IP rule, then W is added to the proof. We'll have more to say about CP and IP shortly.

The proof rules (6.7) will allow us to prove any valid argument involving wffs of the propositional calculus. In fact, we could get by without some of the rules. For example, the contradiction rule (Contr) is not required because the IP rule could be rephrased as "From $\neg A$, derive B and $\neg B$ for some wff B." But we include Contr because it helps make indirect proofs easier to understand and closer to the way we reason informally. Some rules (i.e., DN and the extra rules in DS and Add) can be derived from the other rules. But we include them because they clarify how the rules can be used.

6.3.2 Proofs

Now that we have some proof rules, we can start doing proofs. As we said earlier, a proof (or derivation) is a finite sequence of wffs, where each wff is either a premise or the result of applying a rule to certain previous wffs in the sequence. Note that there are some restrictions on which wffs can be used by the proof rules. We'll get to them when we discuss the CP and IP rules.

We'll write proofs in table format, where each line is numbered and contains a wff together with the reason it's there. For example, a proof sequence

$$W_1, \ldots, W_n$$

will be written in the following form:

$$\begin{array}{lll} 1. & W_1 & \text{Reason for } W_1 \\ 2. & W_2 & \text{Reason for } W_2 \\ \vdots & \vdots & \\ n. & W_n & \text{Reason for } W_n \end{array}$$

The reason column for each line always contains a short indication of why the wff is on the line. If the line depends on previous lines because of a rule, then we'll always include the line numbers of those previous lines and the name of the rule.

Notation for Premises

For each line that contains a premise, we'll write the letter P in the reason column of the line.

example 6.9 A Formal Proof

Earlier we considered the argument with premises $A \vee B$, $\neg A$, and $B \rightarrow C$, and conclusion $B \wedge C$. We "proved" that the argument is valid by saying that the truth of the three premises implies the truth of the conclusion, which can be verified by a truth table. Our goal now is to give a formal proof that the argument is valid by using the rules of our proof system.

We start the proof by writing down the premises on the first three lines and then leaving some space before writing the conclusion as follows:

1.	$A \vee B$	P
2.	$\neg A$	P
3.	$B \to C$	P
	...	
	$B \wedge C$	?

After examining the rules in our proof system, we see that no rule can be used to infer the conclusion from the premises. So a proof will have to include one or more new wffs inserted between the premises and the conclusion in such a way that each wff is obtained by a rule.

How do we proceed? It's a good idea to examine the conclusion to see what might be needed to obtain it. For example, we might observe that the conclusion $B \wedge C$ is the conjunction of B and C. So if we can somehow get B and C on two lines, then the Conj rule would give us the conclusion $B \wedge C$. Then we might notice that the DS rule applied to the premises $A \vee B$ and $\neg A$ gives us B. Finally, we might see that the MP rule applied to B and the premise $B \to C$ gives us C. Here's a finished proof.

1.	$A \vee B$	P
2.	$\neg A$	P
3.	$B \to C$	P
4.	B	1, 2, DS
5.	C	3, 4, MP
6.	$B \wedge C$	4, 5, Conj
	QED.	

end example

Proofs with CP or IP

Suppose a proof consists of a derivation from the premise A to B followed by the application of CP to obtain $A \rightarrow B$. If no other lines of the proof use CP or IP, then the derivation from the premise A to B uses only rules that are valid arguments. So the truth of A implies the truth of B. Therefore, $A \rightarrow B$ is a tautology. For IP we get a similar result. Suppose a proof consists of a derivation from the premise $\neg W$ to False followed by the application of IP to obtain W. If no other lines of the proof use CP or IP, then the derivation from $\neg W$ to False uses only rules that are valid arguments. So the truth of $\neg W$ leads to a contradiction. Therefore, W is a tautology.

When CP is applied to a derivation from the premise A to B to obtain $A \rightarrow B$, the premise A is said to be *discharged*. Similarly, when IP is applied to a derivation from the premise $\neg W$ to False to obtain W, the premise $\neg W$ is said to be *discharged*.

When proofs contain more than one application of CP or IP, then there are restrictions on which lines of a proof can be used to justify later lines. Basically, if a premise has been discharged by the use of CP or IP, then the premise together with the wffs derived from it may not be used to justify any subsequent lines of the proof. We'll get to that shortly. For now, we'll look at some simple examples that have just one application of CP at the end of the proof.

Multiple Premises for CP

The condition for the CP rule is a derivation from a premise A to B. If A is a conjunction of wffs, such as $A = C \wedge D$, then we usually need to work with C and D separately. We can do this by using Simp as follows:

1.	$C \wedge D$	P
2.	C	1, Simp
3.	D	2, Simp

But we can save time and space by writing C and D as premises to begin with as follows:

1.	C	P
2.	D	P

So any derivation that results in B in one proof can also be done in the other proof. If this is the case, then CP can be applied to obtain $C \wedge D \rightarrow B$. In the first proof CP discharges the premise $C \wedge D$ and in the second proof CP discharges the two premises C and D.

The last line of a CP or IP Proof

If a proof ends by applying CP or IP to a derivation, then instead of writing the result of CP or IP on the last line, we will write QED along with the line numbers of the derivation followed by CP or IP. We omit the writing of the result because it can be quite lengthy in some cases. The next examples demonstrate the idea.

example 6.10 A CP Proof

We'll prove that the following conditional wff is a tautology.

$$(A \vee B) \wedge \neg A \wedge (B \rightarrow C) \rightarrow (B \wedge C).$$

Since the antecedent of the wff is a conjunction of three wffs, it suffices to construct a proof of $B \wedge C$ from the three premises $A \vee B$, $\neg A$, and $B \rightarrow C$, and then apply the CP rule. Here's a proof.

1.	$A \vee B$	P
2.	$\neg A$	P
3.	$B \rightarrow C$	P
4.	B	2, 3, DS
5.	C	4, 5, MP
6.	$B \wedge C$	5, 6, Conj
	QED	1–7, CP.

The last line of the proof indicates that CP has been applied to the derivation on lines 1 thru 6 that starts with the three premises and ends with $B \wedge C$. Therefore, the result of CP applied to the derivation is the wff

$$(A \vee B) \wedge \neg A \wedge (B \to C) \to (B \wedge C).$$

Our convention is to not write out the wff, but rather to put QED in its place. Since the only use of CP or IP is at the end of the proof, it follows that the wff is a tautology.

end example

example 6.11 A Simple CP Proof of $A \to A$

We'll prove the simple tautology $A \to A$. In other words, we'll show that A follows from the premise A.

1.	A	P
2.	$A \vee A$	1, Add
3.	$A \wedge (A \vee A)$	1, 2, Conj
4.	A	2, Simp
	QED	1–4, CP.

end example

Subproofs and Discharged Premises

It is often the case that a proof contains another proof, called a *subproof*, which proves a statement that is needed for the proof to continue. A subproof is a derivation that always starts with a new premise and always ends by applying CP or IP to the derivation. When a subproof ends, the premise is discharged and the wffs of the derivation become inactive, which means that they may not be used to justify any subsequent line of the proof.

A subproof is denoted by indenting the wffs of the derivation. We do not indent the wff on the CP or IP line because it is needed for the proof to continue. Here's an example of a proof structure that contains a subproof.

1.	W_1		P
2.	W_2		Reason (May use line 1)
3.	W_3		Reason (May use lines 1–2)
4.		W_4	P (New premise for subproof)
5.		W_5	Reason (May use lines 1–4)
6.		W_6	Reason (May use lines 1–5)
7.		W_7	Reason (May use lines 1–6)
8.	W_8		CP or IP from lines 4–7
9.	W_9		Reason (May use lines 1–3 and 8)
10.	W_{10}		Reason (May use lines 1–3 and 8–9)
	QED		CP or IP from lines 1–3 and 8–10.

The premise on line 4 is discharged by the use of CP or IP on line 8. So lines 4 thru 7 become inactive and may not be used to justify subsequent lines of the proof.

If a subproof is nested within another subproof, then the premise of the innermost subproof must be discharged before the premise of the outermost subproof. In other words, finish the innermost subproof first with CP or IP so the result is available for use as part of the outermost subproof.

Subproofs usually depend on prior premises that have not yet been discharged. For example, suppose that we have a derivation from the premise A to B and we apply CP to obtain $A \to B$. Suppose further that the derivation from A to B depends on a prior premise C that is not discharged. Then B depends on both C and A. In other words, the truth of C and A implies the truth of B. So we have the tautology

$$(C \wedge A) \to B.$$

It follows from Exercise 9c of Section 6.2 that

$$(C \wedge A) \to B \equiv C \to (A \to B).$$

Since C is not discharged, we can apply MP to C and $C \to (A \to B)$ to obtain $A \to B$. This result generalizes to the case where a subproof depends on two or more prior undischarged premises.

So we've found that if a derivation from the premise A to B depends on prior undischarged premises, then the truth of $A \rightarrow B$ depends on the truth of those prior premises. This means that $A \rightarrow B$ is the conclusion of a valid argument from those prior premises. So $A \rightarrow B$ can be used to justify subsequent lines of the proof until one of those prior premises becomes discharged.

The argument is the same for the IP rule. In other words, if a derivation from the premise $\neg W$ to False depends on prior undischarged premises, then W is the conclusion of a valid argument from those premises. So W can be used to justify subsequent lines of the proof until one of those prior premises becomes discharged.

It follows from the preceding paragraphs that if a proof satisfies the restrictions we've described and all premises in the proof are discharged, then the last wff of the proof is a tautology. Here's a listing of the restrictions with CP and IP.

Restrictions with CP and IP

After CP or IP has been applied to a derivation, the lines of the derivation become inactive and may not be used to justify subsequent lines of the proof.

For nested subproofs, finish the innermost subproof first with CP or IP so the result is available for use as part of the outermost subproof.

Let's do some examples to cement the ideas. We'll start with some CP examples and then introduce the use of IP.

example 6.12 A Proof with a Subproof

We'll give a proof to show that the following wff is a tautology.

$$((A \vee B) \rightarrow (B \wedge C)) \rightarrow (B \rightarrow C).$$

This wff is a conditional and the conclusion is also a conditional. So the proof will contain a subproof of the conditional $B \rightarrow C$.

1.	$(A \vee B) \to (B \wedge C)$		P
2.		B	P [for $B \to C$]
3.		$A \vee B$	2, Add
4.		$B \wedge C$	1, 3, MP
5.		C	4, Simp
6.	$B \to C$		2–5, CP
	QED		1, 6, CP.

end example

example 6.13 A Simple Tautology: $A \to (B \to A)$

We'll prove the simple tautology $A \to (B \to A)$.

1.	A		P
2.		B	P [for $B \to A$]
3.		$A \wedge B$	1, 2, Conj
4.		A	3, Simp
5.	$B \to A$		2–4, CP
	QED		1, 5, CP.

end example

example 6.14 Another Simple Tautology: $\neg A \to (A \to B)$

We'll prove the simple tautology $\neg A \to (A \to B)$.

1.	$\neg A$	P
2.	$\quad A$	P [for $A \to B$]
3.	$\quad A \vee B$	2, Add
4.	$\quad B$	1, 3, DS
5.	$A \to B$	2–4, CP
	QED	1, 5, CP.

end example

example 6.15 A Proof with a Subproof with a Subproof

We'll give a proof that the following wff is a tautology.

$$(A \to (B \to C)) \to ((A \to B) \to (A \to C)).$$

This wff is a conditional and its conclusion is also a conditional with a conditional in its conclusion. So the proof will contain a subproof of $(A \to B) \to (A \to C)$, which will start with $A \to B$ as a new premise. This subproof will contain a subproof of $A \to C$, which will start with A as a new premise. Here's the proof.

1.	$A \to (B \to C)$	P
2.	$\quad A \to B$	P [for $(A \to B) \to (A \to C)$]
3.	$\quad\quad A$	P [for $A \to C$]
4.	$\quad\quad B \to C$	1, 3, MP
5.	$\quad\quad B$	2, 3, MP
6.	$\quad\quad C$	4, 5, MP
7.	$\quad A \to C$	3–6, CP
8.	$(A \to B) \to (A \to C)$	2, 7, CP
	QED	1, 8, CP.

end example

Using the IP Rule

Suppose we want to prove a statement, but we just can't seem to find a way to get going. We might try *proof by contradiction* (i.e., *reductio ad absurdum*). In other words, assume that the statement to be proven is false and argue from there until a contradiction of some kind is reached. Then conclude that the original statement is true. The idea is based on the following simple equivalence for any wff W:

$$W \equiv \neg W \to \text{False}.$$

This equivalence together with the CP rule gives us the IP rule. Here's why. If we have a derivation from the premise $\neg W$ to False, then we can apply CP to obtain $\neg W \to$ False, which is equivalent to W. So we have the IP rule.

Since False is equivalent to $B \wedge \neg B$ for any wff B, there can be different ways to find a contradiction. You might try proof by contradiction whenever there doesn't seem to be enough information from the given premises or when you run out of ideas.

Using IP to Prove a Conditional

When proving a conditional of the form $V \to W$, it is often easier to use IP as part of a CP proof as follows: Start the CP proof by listing the antecedent V as a premise for the CP proof. Then start an IP subproof by listing $\neg W$ as a new premise. Once a contradiction is reached the IP rule can be used to infer W. Then CP gives the desired result.

example 6.16 A CP Proof with an IP Subproof

We'll prove that the following conditional is a tautology.

$$(A \vee B) \wedge (B \to C) \vee \neg C \to A.$$

The antecedent is a conjunction of the three wffs $A \vee B$, $B \to C$, and $\neg C$. So we'll start the proof by writing down these three wffs as premises for CP. The goal is to obtain A and apply CP. We'll obtain A with an IP subproof by assuming $\neg A$ as a new premise.

1.	$A \vee B$		P
2.	$B \to C$		P
3.	$\neg C$		P
4.		$\neg A$	P [for A]
5.		B	1, 4, DS
6.		C	2, 5, MP
7.		False	3, 6, Contr
8.	A		4–7, IP
	QED		1–3, 6, CP.

end example

example 6.17 An IP Proof with an IP Subproof of $A \vee \neg A$

We'll give an IP proof of the simple tautology

$$A \vee \neg A.$$

The proof starts with the negation of the wff as a new premise. Then in an attempt to find a contradiction, it starts an IP subproof to obtain A by assuming $\neg A$.

1.	$\neg (A \vee \neg A)$		P [for $A \vee \neg A$]
2.		$\neg A$	P [for A]
3.		$A \vee \neg A$	2, Add
4.		False	1, 3, Contr
5.	A		2–4, IP
6.	$A \vee \neg A$		5, Add
7.	False		1, 6, Contr
	QED		1, 5–7, IP.

end example

example 6.18 Using IP Twice with Descartes

We'll prove the validity of the following argument noted earlier, which might have been made by Descartes.

If I think, then I exist.
If I do not think, then I think.
Therefore, I exist.

We can formalize the argument by letting A and B be "I think" and "I exist," respectively. Then the argument can be represented by saying that the conclusion B follows from the premises $A \to B$ and $\neg A \to A$. Here's a proof.

1.	$A \to B$			P
2.	$\neg A \to A$			P
3.		$\neg B$		P[for B]
4.			$\neg \neg A$	P [for $\neg A$]
5.			A	4, DN
6.			B	1, 5, MP
7.			False	3, 6, Contr
8.		$\neg A$		4–7, IP
9.		A		2, 8, MP
10.		False		8, 9, Contr
11.	B			3, 8–10, IP
	QED.			

end example

6.3.3 Derived Rules

There are other useful rules that can be derived from the rules in our list (6.7). In fact, some rules in the list can be derived from the other rules in the list (i.e., DN and the additional rules in DS and Add). But we include them because they make the rules easier to use. The next three examples show how to derive these three rules. After that, we'll derive four new proof rules that have many uses.

example 6.19 The Double Negation (DN) Rules

We'll show that the double negation rules can be derived from the other original rules. In other words, we'll derive the following rules.

$$\frac{\neg \neg A}{A} \text{ and } \frac{A}{\neg \neg A}.$$

Here's a proof that the first rule represents a valid argument.

1.	$\neg\neg A$		P
2.		$\neg A$	P [for A]
3.		False	1, 2, Contr
4.	A		2, 3, IP
	QED.		

Here's a proof that the second rule represents a vaild argument.

1.	A		P
2.		$\neg\neg\neg A$	P [for $\neg\neg A$]
3.		$\neg A$	2, DN [proved above]
4.		False	1, 3, Contr
5.	$\neg\neg A$		2–4, IP
	QED.		

end example

example 6.20 The Other Disjunctive Syllogism (DS) Rule

We'll show that one disjunctive syllogism rule can be derived from the other original rules. In other words, we'll derive the following rule.

$$\frac{A \vee B,\ \neg B}{A}.$$

Here's a proof that the rule represents a valid argument.

1.	$A \vee B$		P
2.	$\neg B$		P
3.		$\neg A$	P [for A]
4.		B	1, 3, DS [other DS]
5.		False	2, 4, Contr
6.	A		3–5, IP
	QED.		

end example

example 6.21 The Other Addition (Add) Rule

We'll show that one addition rule can be derived from the other original rules. In other words, we'll derive the following rule.

$$\frac{A}{B \vee A}.$$

Here's a proof that the rule represents a valid argument.

1.	A			P
2.	$A \vee B$			1, Add
3.		$\neg (B \vee A)$		P [for $B \vee A$]
4.			$\neg B$	P [for B]
5.			$\neg A$	2, 4, DS
6.			False	1, 5, Contr
7.		B		4–6, IP
8.		$B \vee A$		7, Add [other Add]
9.		False		3, 8, Contr
10.	$B \vee A$			3, 7–9, IP
	QED.			

end example

Four New Derived Rules

Here are four derived rules that can be very useful. We'll list them next and then give proofs that they are derivable from the original rules.

Derived Rules **(6.8)**

Modus Tollens (MT) (Latin for "mode that denies") $\dfrac{A \to B, \quad \neg B}{\neg A}.$	*Proof by Cases (Cases)* $\dfrac{A \vee B, \quad A \to C, \quad B \to C}{C}.$
Hypothetical Syllogism (HS) $\dfrac{A \to B, \quad B \to C}{A \to C}.$	*Constructive Dilemma (CD)* $\dfrac{A \vee B, \quad A \to C, \quad B \to D}{C \vee D}.$

example 6.22 Proof of Modus Tollens (MT)

We'll show that the modus tollens rule can be derived from the original rules. In other words, we'll derive the following rule.

$$\frac{A \to B, \quad \neg B}{\neg A}.$$

Here's a proof that the rule represents a valid argument.

1.	$A \to B$		P
2.	$\neg B$		P
3.		$\neg\neg A$	P [for $\neg$ A]
4.		A	3, DN
5.		B	1, 4, MP
6.		False	2, 5, Contr
7.	$\neg A$		3–6, IP
	QED.		

end example

example 6.23 Proof of Proof by Cases (Cases)

We'll show that the proof by cases rule can be derived from the original rules. In other words, we'll derive the following rule.

$$\frac{A \vee B, \quad A \to C, \quad B \to C}{C}.$$

Here's a proof that the rule represents a valid argument.

1.	$A \vee B$		P
2.	$A \to C$		P
3.	$B \to C$		P
4.		$\neg C$	P [for C]
5.		$\neg A$	4, MT
6.		B	1, 5, DS
7.		C	3, 6, MP
8.		False	4, 7, Contr
9.	C		4–8, IP
	QED.		

end example

example 6.24 Proof of Hypothetical Syllogism (HS)

We'll show that the hypothetical syllogism rule can be derived from the original rules. In other words, we'll derive the following rule.

$$\frac{A \to B, \quad B \to C}{A \to C}.$$

Here's a proof that the rule represents a valid argument.

1.	$A \to B$	P
2.	$B \to C$	P
3.	A	P [for $A \to C$]
4.	B	1, 3, MP
5.	C	2, 4, MP
6.	$A \to C$	3–5, CP
	QED.	

end example

example 6.25 Proof of Constructive Dilemma (CD)

We'll show that the constructive dilemma rule can be derived from the original rules. In other words, we'll derive the following rule.

$$\frac{A \vee B, \quad A \to C, \quad B \to D}{C \vee D}.$$

Here's a proof that the rule represents a valid argument.

1.	$A \vee B$	P
2.	$A \to C$	P
3.	$B \to D$	P
4.	A	P [for $A \to C \vee D$]
5.	C	2, 4, MP
6.	$C \vee D$	5, Add
7.	$A \to C \vee D$	4–6, CP
8.	B	P [for $B \to C \vee D$]
9.	D	3, 8, MP
10.	$C \vee D$	9, Add
11.	$B \to C \vee D$	8–10, CP
12.	$C \vee D$	1, 7, 11, Cases
	QED.	

end example

6.3.4 Theorems, Soundness, and Completeness

We'll begin by giving the definition of a theorem for our proof system. Then we'll discuss how theorems are related to tautologies.

> **Definition of Theorem**
>
> A *theorem* is the last wff in a proof for which all premises have been discharged.

A proof system for propositional calculus is said to be *sound* if every theorem is a tautology. It's nice to know that our proof system is sound. This follows from our previous discussion about the fact that the last wff of a proof is a tautology if all premises have been discharged.

A proof system for propositional calculus is said to be *complete* if every tautology is a theorem. It's nice to know that our proof system is complete. This fact follows from a completeness result in the next section for a system that depends on three wffs that it uses as axioms. The completeness of our system follows because each of three wffs used as axioms has a proof in our system.

Using Previously Proved Results (Theorems)

If we know that some wff is a theorem, then we can use it in another proof. That is, we can place the theorem on some line of the proof. If the theorem is a conditional of the form $V \to W$, then we can use it as a derived proof rule. In other words, if we find V on some line of a proof, we can write W on a subsequent line.

The proof rules can be used to prove the basic equivalences listed in (6.1). So each equivalence $V \equiv W$ gives us two derived rules, one for $V \to W$ and the other for $V \to W$.

When we place a theorem on a proof line or when we use a theorem as a derived proof rule, we'll indicate it by writing

$$T$$

in the reason column. Think of T as a theorem.

example 6.26 Proving a Theorem Using Three Theorems

We'll prove the simple tautology $(A \to B) \vee (B \to A)$ by using three theorems and the constructive dilemma rule.

1.	$A \vee \neg A$	T [Example 6.17]
2.	$A \to (B \to A)$	T [Example 6.13]
3.	$\neg A \to (A \to B)$	T [Example 6.14]
4.	$(A \to B) \vee (B \to A)$	1, 2, 3, CD
	QED.	

end example

6.3.5 Practice Makes Perfect

Some proofs are straightforward, while others can be brain busters. Remember, when you construct a proof, it may take several false starts before you come up with a correct proof sequence. Study the examples and then do lots of exercises.

Basic Equivalences

A good way to practice formal proof techniques is to write proofs for the basic equivalences. We've already seen proofs for the tautologies $A \to A$ and $A \vee \neg A$ in Example 6.11 and Example 6.17, respectively. In the following examples we'll give proofs for a few more of the basic equivalences. Proofs for other basic equivalences are in the exercises.

example 6.27 $A \vee B \equiv B \vee A$

Proof of $A \vee B \to B \vee A$:

1.	$A \vee B$			P
2.		$\neg (B \vee A)$		P [for $B \vee A$]
3.			$\neg \neg A$	P [for $\neg A$]
4.			A	3, DN
5.			$B \vee A$	4, Add
6.			False	2, 5, Contr
7.		$\neg A$		3–6, IP
8.		B		1, 7, DS
9.		$B \vee A$		8, Add
10.		False		2, 9, Contr
11.	$B \vee A$			2, 7–10, IP
	QED			1, 11, CP.

The proof of $B \vee A \to A \vee B$ is similar.

end example

example 6.28 $A \rightarrow B \equiv \neg A \vee B$

Proof of $(A \rightarrow B) \rightarrow (\neg A \vee B)$:

1.	$A \rightarrow B$	P
2.	$\quad \neg(\neg A \vee B)$	P [for $\neg A \vee B$]
3.	$\qquad \neg A$	P [for A]
4.	$\qquad \neg A \vee B$	3, Add
5.	$\qquad$ False	2, 4, Contr
6.	$\quad A$	3–5, IP
7.	$\quad B$	1, 6, MP
8.	$\quad \neg A \vee B$	7, Add
9.	$\quad$ False	2, 8, Contr
10.	$\neg A \vee B$	2, 6–9, IP
	QED	1, 10, CP.

Proof of $(\neg A \vee B) \rightarrow (A \rightarrow B)$:

1.	$\neg A \vee B$	P
2.	$\quad A$	P [for $A \rightarrow B$]
3.	$\quad \neg\neg A$	2, DN
4.	$\quad B$	1, 3, DS
5.	$A \rightarrow B$	2–4, CP
	QED	1, 5, CP.

end example

example 6.29 $\neg(A \rightarrow B) \equiv A \wedge \neg B$

Proof of $\neg (A \to B) \to (A \wedge \neg B)$:

1.	$\neg (A \to B)$	P
2.	$\quad \neg A$	P [for A]
3.	$\quad \neg A \vee B$	2, Add
4.	$\quad A \to B$	3, T [Example 6.28]
5.	$\quad$ False	1, 4, Contr
6.	A	2–5, IP
7.	$\quad \neg\neg B$	P [for $\neg B$]
8.	$\quad B$	7, DN
9.	$\quad B \to (A \to B)$	T, [Example 6. 13]
10.	$\quad A \to B$	8, 9, MP
11.	$\quad$ False	1, 10, Contr
12.	$\neg B$	7–11, IP
13.	$A \wedge \neg B$	6, 12, Conj
	QED	1, 6, 12, 13, CP.

Proof of $(A \wedge \neg B) \to \neg (A \to B)$:

1.	$A \wedge \neg B$	P
2.	$\quad \neg\neg (A \to B)$	P [for $\neg (A \to B)$]
3.	$\quad A \to B$	2, DN
4.	$\quad A$	1, Simp
5.	$\quad B$	3, 4, MP
6.	$\quad \neg B$	1, Simp
7.	$\quad$ False	5, 6, Contr
8.	$\neg (A \to B)$	2–7, IP
	QED	1, 6, CP.

end example

example 6.30 $\neg (A \vee B) \equiv \neg A \wedge \neg B$

Proof of $\neg (A \vee B) \to (\neg A \wedge \neg B)$:

1.	$\neg (A \vee B)$		P
2.		$\neg\neg A$	P [for $\neg A$]
3.		A	2, DN
4.		$A \vee B$	3, Add
5.		False	1, 4, Contr
6.	$\neg A$		2–5, IP
7.		$\neg\neg B$	P [for $\neg B$]
8.		B	7, DN
9.		$A \vee B$	8, Add
10.		False	1, 9, Contr
11.	$\neg B$		7–10, IP
12.	$\neg A \wedge \neg B$		6, 11, Conj
	QED		1, 6, 11, 12, CP.

Proof of $(\neg A \wedge \neg B) \to \neg (A \vee B)$:

1.	$\neg A \wedge \neg B$	P
2.	$\neg\neg(A \vee B)$	P [for $\neg(A \vee B)$]
3.	$A \vee B$	2, DN
4.	$\neg A$	1, Simp
5.	B	3, 4, DS
6.	$\neg B$	1, Simp
7.	False	5, 6, Contr
8.	$\neg(A \vee B)$	2–7, IP
	QED	1, 8, CP.

end example

Exercises

Proof Structures

1. Let W denote the wff $(A \vee B \to C) \to (B \vee C)$. It's easy to see that W is not a tautology. For example, let $A = B = C =$ False. Suppose someone claims that the following proof shows that W is a tautology.

1.	$A \vee B \to C$	P
2.	A	P
3.	$A \vee B$	2, Add
4.	C	1, 3, MP
5.	$B \vee C$	4, Add
	QED	1–5, CP.

What is wrong with the claim?

2. Let W denote the wff $(A \to (B \wedge C)) \to (A \to B) \wedge C$. It's easy to see that W is not a tautology. Suppose someone claims that the following sequence of statements is a "proof" of W:

1.	$A \to (B \wedge C)$	P
2.	A	P [for $A \to B$]
3.	$B \wedge C$	1, 2, MP
4.	B	3, Simp
5.	$A \to B$	2–4, CP
6.	C	3, Simp
7.	$(A \to B) \wedge C$	5, 6, Conj
	QED	1, 5–7, CP.

What is wrong with this "proof" of W?

3. Find the number of premises required for a proof of each of the following wffs. Assume that the letters stand for other wffs.

a. $A \to (B \to (C \to D))$.
b. $((A \to B) \to C) \to D$.

4. Give a formalized version of the following proof.

If I am dancing, then I am happy. There is a mouse in the house or I am happy. I am sad. Therefore, there is a mouse in the house and I am not dancing.

Formal Proofs

5. Give a formal proof for each of the following tautologies by using the CP rule. Do not use the IP rule.

a. $A \to (B \to (A \wedge B))$.
b. $A \to (\neg B \to (A \wedge \neg B))$.
c. $(A \vee B \to C) \wedge A \to C$.
d. $(B \to C) \to (A \wedge B \to A \wedge C)$.
e. $(A \vee B \to C \wedge D) \to (B \to D)$.
f. $(A \vee B \to C) \wedge (C \to D \wedge E) \to (A \to D)$.
g. $(\neg A \vee \neg B) \wedge (B \vee C) \wedge (C \to D) \to (A \to D)$.
h. $(A \to (B \to C)) \to (B \to (A \to C))$.
i. $(A \to C) \to (A \wedge B \to C)$.

j. $(A \to C) \to (A \to B \vee C)$.
k. $(A \to B) \to (C \vee A \to C \vee B)$.

6. Give a formal proof for each of the following tautologies by using the CP rule and by using the IP rule at least once in each proof.

a. $A \to (B \to A)$.
b. $(A \to B) \wedge (A \vee B) \to B$.
c. $\neg B \to (B \to C)$.
d. $(A \to C) \to (A \to B \vee C)$.
e. $(A \to B) \to ((A \to \neg B) \to \neg A)$.
f. $(A \to B) \to ((B \to C) \to (A \vee B \to C))$.
g. $(A \to B) \to (C \vee A \to C \vee B)$.
h. $(C \to A) \wedge (\neg C \to B) \to (A \vee B)$.

7. Give a formal proof for each of the following tautologies by using the CP rule and by using the IP rule at least once in each proof.

a. $(A \vee B \to C) \wedge A \to C$.
b. $(B \to C) \to (A \wedge B \to A \wedge C)$.
c. $(A \vee B \to C \wedge D) \to (B \to D)$.
d. $(A \vee B \to C) \wedge (C \to D \wedge E) \to (A \to D)$.
e. $\neg (A \wedge B) \wedge (B \vee C) \wedge (C \to D) \to (A \to D)$.
f. $(A \to B) \to ((B \to C) \to (A \vee B \to C))$.
g. $(A \to (B \to C)) \to (B \to (A \to C))$.
h. $(A \to C) \to (A \wedge B \to C)$.

Challenges

8. Prove that the following rule, called the Destructive Dilemma rule, can be derived from the original and derived proof rules.

$$\frac{\neg C \vee \neg D, \quad A \to C, \quad B \to D}{\neg A \vee \neg B}.$$

9. Two students came up with the following different wffs to formalize the statement "If A then B else C."

$(A \wedge B) \vee (\neg A \wedge C)$.

$(A \rightarrow B) \wedge (\neg A \rightarrow C)$.

Prove that the two wffs are equivalent by finding formal proofs for the following two statements.

a. $((A \wedge B) \vee (\neg A \wedge C)) \rightarrow ((A \rightarrow B) \wedge (\neg A \rightarrow C))$.

b. $((A \rightarrow B) \wedge (\neg A \rightarrow C)) \rightarrow ((A \wedge B) \vee (\neg A \wedge C))$.

Basic Properties

For each of the following exercises try to use only the proof rules and derived proof rules. If you must use T, then use it only if the theorem has already been proved.

10. Prove the following basic properties of conjunction.

a. $A \wedge \neg \text{False} \equiv A$.

b. $A \wedge \text{False} \equiv \text{False}$.

c. $A \wedge A \equiv A$.

d. $A \wedge \neg A \equiv \text{False}$.

e. $A \wedge B \equiv B \wedge A$.

f. $A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$.

11. Prove the following basic properties of disjunction.

a. $A \vee \neg \text{False} \equiv \neg \text{False}$.

b. $A \vee \text{False} \equiv A$.

c. $A \vee A \equiv A$.

d. $A \vee (B \vee C) \equiv (A \vee B) \vee C$.

12. Prove the following basic properties of implication.

a. $A \rightarrow \neg \text{False}$.

b. $A \rightarrow \text{False} \equiv \neg A$.

c. $\neg \text{False} \rightarrow A \equiv A$.

d. $\text{False} \rightarrow A$.

13. Prove the following basic conversions.

a. $A \to B \equiv \neg B \to \neg A$.

b. $A \to B \equiv A \wedge \neg B \to \text{False}$.

c. $\neg (A \wedge B) \equiv \neg A \vee \neg B$.

d. $A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$.

e. $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$.

14. Prove the following absorption equivalences.

a. $A \wedge (A \vee B) \equiv A$.

b. $A \vee (A \wedge B) \equiv A$.

c. $A \wedge (\neg A \vee B) \equiv A \wedge B$.

d. $A \vee (\neg A \wedge B) \equiv A \vee B$.

6.4 Formal Axiom Systems

Although truth tables are sufficient to decide the truth of a propositional wff, most of us do not reason by truth tables. We reason in a way that is similar to the natural deduction approach of Section 6.3, where we used the proof rules listed in (6.7) and (6.8). In this section we'll introduce the *axiomatic* approach, where the premises for arguments are limited to a fixed set of wffs, called *axioms*.

Our goal is to have a formal proof system with two properties. We want our proofs to yield theorems that are tautologies, and we want any tautology to be provable as a theorem. In other words, we want soundness and completeness.

Soundness: All proofs yield theorems that are tautologies.

Completeness: All tautologies are provable as theorems.

6.4.1 An Example Axiom System

Is there a simple formal system for which we can show that the propositional calculus is both sound and complete? Yes, there is. In fact, there are many of them. Each one specifies a small fixed set of axioms and inference rules. The pioneer in this area was the mathematician Gottlob Frege (1848–1925). He formulated the first such axiom system [1879]. We'll discuss it further in the exercises. Later, in 1930, J. Lukasiewicz showed that Frege's system, which has six axioms, could be replaced by the following three axioms, where *A, B*, and *C* can represent any wff generated by propositional variables and the two connectives $\neg$ and $\rightarrow$.

Frege-Lukasiewicz Axioms **(6.9)**

1. $A \rightarrow (B \rightarrow A)$.

2. $(A \rightarrow (B \rightarrow C)) \rightarrow ((A \rightarrow B) \rightarrow (A \rightarrow C))$.

3. $(\neg A \rightarrow \neg B) \rightarrow (B \rightarrow A)$.

The only inference rule used is modus ponens. Although the axioms may appear a bit strange, they can all be verified by truth tables. Also note that conjunction and disjunction are missing. But this is no problem, since we know that they can be written in terms of implication and negation.

We'll use this system to prove the CP rule (i. e., the deduction theorem). But first we must prove a result that we'll need, namely that $A \rightarrow A$ is a theorem provable from the axioms. Notice that the proof uses only the given axioms and modus ponens.

Lemma 1. $A \rightarrow A$ is provable from the axioms.

In the following proof *B* can be any wff, including *A*.

1.	$(A \rightarrow ((B \rightarrow A) \rightarrow A)) \rightarrow ((A \rightarrow (B \rightarrow A)) \rightarrow (A \rightarrow A))$	Axiom 2
2.	$A \rightarrow ((B \rightarrow A) \rightarrow A)$	Axiom 1
3.	$(A \rightarrow (B \rightarrow A)) \rightarrow (A \rightarrow A)$	1, 2, MP
4.	$A \rightarrow (B \rightarrow A)$	Axiom 1
5.	$A \rightarrow A$	3, 4, MP
	QED.	

Now we're in position to prove the CP rule, which is also called the deduction theorem. The proof, which is due to Herbrand (1930), uses only the given axioms, modus ponens, and Lemma 1.

Deduction Theorem (Conditional Proof Rule, CP)

If A is a premise in a proof of B, then there is a proof of $A \rightarrow B$ that does not use A as a premise.

Proof: Assume that A is a premise in a proof of B. We must show that there is a proof of $A \rightarrow B$ that does not use A as a premise. Suppose that $B_1, \ldots, B_n$ is a proof of B that uses the premise A. We'll show by induction that for each k in the interval $1 \leq k \leq n$, there is a proof of $A \rightarrow B_k$ that does not use A as a premise. Since $B = B_n$, the result will be proved. If $k = 1$, then either $B_1 = A$ or B_1 is an axiom or a premise other than A. If $B_1 = A$, then $A \rightarrow B_1 = A \rightarrow A$, which by Lemma 1 has a proof that does not use A as a premise. If B_1 is an axiom or a premise other than A, then the following proof of $A \rightarrow B_1$ does not use A as a premise.

1.	B_1	An axiom or premise other than A
2.	$B_1 \rightarrow (A \rightarrow B_1)$	Axiom 1
3.	$A \rightarrow B_1$	1, 2, MP

Now assume that for each $i < k$ there is a proof of $A \rightarrow B_i$ that does not use A as a premise. We must show that there is a proof of $A \rightarrow B_k$ that does not use A as a premise. If $B_k = A$ or B_k is an axiom or a premise other than A, then we can use the same argument for the case when $k = 1$ to conclude that there is a proof of $A \rightarrow B_k$ that does not use A as a premise. If B_k is not an axiom or a premise, then it is inferred by MP from two wffs in the proof of the form B_i and $B_j = B_i \rightarrow B_k$, where $i < k$ and $j < k$. Since by $i < k$ and $j < k$, the induction hypothesis tells us that there are proofs of $A \rightarrow B_i$ and $A \rightarrow (B_i \rightarrow B_k)$, neither of which contains A as a premise. Now consider the following proof, where lines 1 and 2 represent the proofs of $A \rightarrow B_i$ and $A \rightarrow (B_i \rightarrow B_k)$.

1.	Proof of $A \rightarrow B_i$	Induction
2.	Proof of $A \rightarrow (B_i \rightarrow B_k)$	Induction
3.	$(A \rightarrow (B_i \rightarrow B_k)) \rightarrow ((A \rightarrow B_i) \rightarrow (A \rightarrow B_k))$	Axiom 2
4.	$(A \rightarrow B_i) \rightarrow (A \rightarrow B_k)$	2, 3, MP
5.	$A \rightarrow B_k$	1, 4, MP

So there is a proof of $A \rightarrow B_k$ that does not contain A as a premise. Let $k = n$ to obtain the desired result because $B_n = B$. QED.

Once we have the CP rule, proofs become much easier since we can have premises in our proofs. But still, everything that we do must be based only on the axioms, MP, CP, and any results we prove along the way. The system is sound because the axioms are tautologies and MP maps tautologies to a tautology. In other words, every proof yields a theorem that is a tautology.

The remarkable thing is that this little axiom system is complete in the sense that there is a proof within the system for every tautology of the propositional calculus. We'll give a few more examples to get the flavor of reasoning from a very small set of axioms.

example 6.31 Hypothetical Syllogism

We'll use CP to prove the hypothetical syllogism proof rule.

Hypothetical Syllogism (HS) **(6.10)**

From the premises $A \rightarrow B$ and $B \rightarrow C$, there is a proof of $A \rightarrow C$.

Proof:	1.	$A \to B$	P
	2.	$B \to C$	P
	3.	A	P [for $A \to C$]
	4.	B	1, 3, MP
	5.	C	2, 4, MP
	6.	$A \to C$	3–5, CP
		QED.	

We can apply the CP rule to HS to say that from the premise $A \to B$ there is a proof of $(B \to C) \to (A \to C)$. One more application of the CP rule tells us that the following wff has a proof with no premises:

$$(A \to B) \to ((B \to C) \to (A \to C)).$$

This is just another way to represent the HS rule as a tautology.

end example

Six Sample Proofs

Now that we have CP and HS rules, it should be easier to prove statements within the axiom system. Let's prove the following six statements.

Six Tautologies **(6.11)**

a. $\neg A \to (A \to B)$.

b. $\neg\neg A \to A$.

c. $A \to \neg\neg A$.

d. $(A \to B) \to (\neg B \to \neg A)$.

e. $A \to (\neg B \to \neg(A \to B))$.

f. $(A \to B) \to ((\neg A \to B) \to B)$.

In the next six examples we'll prove these six statements using only the axioms, MP, CP, HS, and previously proven results.

example 6.32 $\neg A \to (A \to B)$

Proof:	1.	$\neg A$	P
	2.	$\neg A \to (\neg B \to \neg A)$	Axiom 1
	3.	$\neg B \to \neg A$	1, 2, MP
	4.	$(\neg B \to \neg A) \to (A \to B)$	Axiom 3
	5.	$A \to B$	3, 4, MP
		QED	1–5, CP.

end example

example 6.33 $\neg \neg A \to A$

Proof:	1.	$\neg \neg A$	P
	2.	$\neg \neg A \to (\neg A \to \neg \neg \neg A)$	T [Example 6.32]
	3.	$\neg A \to \neg \neg \neg A$	1, 2, MP
	4.	$(\neg A \to \neg \neg \neg A) \to (\neg \neg A \to A)$	Axiom 3
	5.	$\neg \neg A \to A$	3, 4, MP
	6.	A	1, 5, MP
		QED	1–6, CP.

end example

example 6.34 $A \to \neg \neg A$

Proof:	1.	$\neg\neg\neg A \to \neg A$	T [Example 6.33]
	2.	$(\neg\neg\neg A \to \neg A) \to (A \to \neg\neg A)$	Axiom 3
	3.	$A \to \neg\neg A$	1, 2, MP
		QED.	

end example

example **6.35** $(A \to B) \to (\neg B \to \neg A)$

Proof:	1.	$A \to B$	P
	2.	$\neg\neg A \to A$	T [Example 6.33]
	3.	$\neg\neg A \to B$	1, 2, HS
	4.	$B \to \neg\neg B$	T [Example 6.34]
	5.	$\neg\neg A \to \neg\neg B$	3, 4, HS
	6.	$(\neg\neg A \to \neg\neg B) \to (\neg B \to \neg A)$	Axiom 3
	7.	$\neg B \to \neg A$	5, 6, MP
		QED	1–7, CP.

end example

example **6.36** $A \to (\neg B \to \neg(A \to B))$

Proof:	1.	A	P
	2.	$\quad A \to B$	P [for $(A \to B) \to B$]
	3.	$\quad B$	1, 2, MP
	4.	$(A \to B) \to B$	2, 3, CP
	5.	$((A \to B) \to B) \to (\neg B \to \neg(A \to B))$	T [Example 6.35]
	6.	$\neg B \to \neg(A \to B)$	4, 5, MP
		QED	1, 4–6, CP.

end example

example **6.37** $(A \to B) \to ((\neg A \to B) \to B)$

Proof:	1.	$A \to B$	P
	2.	$(A \to B) \to (\neg B \to \neg A)$	T [Example 6.35]
	3.	$\neg B \to \neg A$	1, 2, MP
	4.	$\quad \neg A \to B$	P [for $(\neg A \to B) \to B$]
	5.	$\quad \neg B \to B$	3, 4, HS
	6.	$\quad \neg B \to (B \to \neg (A \to B))$	T [Example 6.32]
	7.	$\quad (\neg B \to (B \to \neg (A \to B))) \to ((\neg B \to B) \to (\neg B \to \neg (A \to B)))$	Axiom 2
	8.	$\quad (\neg B \to B) \to (\neg B \to \neg (A \to B))$	6, 7, MP
	9.	$\quad \neg B \to \neg (A \to B)$	5, 8, MP
	10.	$\quad (\neg B \to \neg (A \to B)) \to ((A \to B) \to B))$	Axiom 3
	11.	$\quad (A \to B) \to B)$	9, 10, MP
	12.	$\quad B$	1, 11, MP
	13.	$(\neg A \to B) \to B$	4–12, CP
		QED	1–3, 13, CP.

end example

Completeness of the Axiom System

As we mentioned earlier, it is a remarkable result that this axiom system is complete. The interesting thing is that we now have enough tools to find a proof of completeness.

Lemma 2

Let W be a wff with propositional variables $P_1, \ldots, P_m$. For any truth assignment to the variables, let $Q_1, \ldots, Q_m$ be defined by letting each Q_k be either P_k or $\neg P_k$ depending on whether P_k is assigned True or False, respectively. Then from the premises $Q_1, \ldots, Q_m$ there is either a proof of W or a proof of $\neg W$ depending on whether the assignment makes W true or false, respectively.

Proof: The proof is by induction on the number n of connectives that occcur in W. If $n = 0$, then W is just a propositional variable P. If P is assigned True, then we must find a proof of P using P as its own premise.

1.	P	Premise
2.	$P \to P$	Lemma 1
3.	P	1, 2, MP
	QED	1–3, CP.

If P is assigned False, then we must find a proof of $\neg P$ from premise $\neg P$.

1.	$\neg P$	Premise
2.	$\neg P \to \neg P$	Lemma 1
3.	$\neg P$	1, 2, MP
	QED	1–3, CP.

So assume W is a wff with n connectives where $n > 0$ and assume that the lemma is true for all wffs with less than n connectives. Now W has one of two forms, $W = \neg A$ or $W = A \to B$. It follows that A and B each have less than n connectives, so by induction there are proofs from the premises $Q_1, \ldots, Q_m$ of either A or $\neg A$ and of either B or $\neg B$ depending on whether they are made true or false by the truth assignment. We'll argue for each form of W.

Let $W = \neg A$. If W is true, then A is false, so there is a proof from the premises of $\neg A = W$. If W is false, then A is true, so there is a proof from the premises of A. Now (6.11c) gives us $A \to \neg \neg A$. So by MP there is a proof from the premises of $\neg \neg A = \neg W$.

Let $W = A \to B$. Assume W is true. Then either A is false or B is true. If A is false, then there is a proof from the premises of $\neg A$. Now (6.11a) gives us $\neg A \to (A \to B)$. So by MP there is a proof from the premises of $(A \to B) = W$. If B is true, then there is a proof from the premises of B. Now Axiom 1 gives us $B \to (A \to B)$. So by MP there is a proof from the premises of $(A \to B) = W$.

Assume W is false. Then A is true and B is false. So there is a proof from the premises of A and there is a proof from the premises of $\neg B$. Now (6.11e) gives us $A \to (\neg B \to \neg (A \to B))$. So by two applications of MP there is a proof from the premises of $\neg (A \to B) = \neg W$. QED.

Theorem (Completeness)

Any tautology can be proven as a theorem in the axiom system.

Proof: Let W be a tautology with propositional variables $P_1, \ldots, P_m$. Since W is always true, it follows from Lemma 2 that for any truth assignment to the propositional variables $P_1, \ldots, P_m$ that occur in W, there is a proof of W from the premises $Q_1, \ldots, Q_m$, where each Q_k is either P_k or $\neg P_k$ according to whether the truth assignment to P_k is True or False, respectively. Now if P_m is assigned True, then $Q_m = P_m$ and if P_m is assigned False, then $Q_m = \neg P_m$. So there are two proofs of W, one with premises $Q_1, \ldots, Q_{m-1}, P_m$ and one with premises $Q_1, \ldots, Q_{m-1}, \neg P_m$. Now apply the CP rule to both proofs to obtain the following two proofs.

A proof from premises $Q_1, \ldots, Q_{m-1}$ of $(P_m \to W)$.

A proof from premises $Q_1, \ldots, Q_{m-1}$ of $(\neg P_m \to W)$.

We combine the two proofs into one proof. Now (6.11f) gives us the following statement, which we add to the proof:

$$(P_m \to W) \to ((\neg P_m \to W) \to W).$$

Now with two applications of MP, we obtain W. So there is a proof of W from premises $Q_1, \ldots, Q_{m-1}$. Now use the same procedure for $m - 1$ more steps to obtain a proof of W with no premises. QED.

6.4.2 Other Axiom Systems

There are many other small axiom systems for the propositional calculus that are sound and complete. For example, Frege's original axiom system consists of the following six axioms together with the modus ponens inference rule, where *A, B*, and *C* can represent any wff generated by propositional variables and the two connectives $\neg$ and $\to$.

Frege's Axioms **(6.12)**

1. $A \to (B \to A)$.
2. $(A \to (B \to C)) \to ((A \to B) \to (A \to C))$.
3. $(A \to (B \to C)) \to (B \to (A \to C))$.
4. $(A \to B) \to (\neg B \to \neg A)$.
5. $\neg\neg A \to A$.
6. $A \to \neg\neg A$.

If we examine the proof of the CP rule, we see that it depends only on Axioms 1 and 2 of (6.9), which are the same as Frege's first two axioms. Also, HS is proved with CP. So we can continue reasoning from these axioms with CP and HS in our tool kit. We'll discuss this system in the exercises.

Another small axiom system, due to Hilbert and Ackermann [1938], consists of the following four axioms together with the modus ponens inference rule, where $A \to B$ is used as an abbreviation for $\neg A \vee B$, and where A, B, and C can represent any wff generated by propositional variables and the two connectives $\neg$ and $\vee$.

Hilbert-Ackermann Axioms **(6.13)**

1. $A \vee A \to A$.
2. $A \to A \vee B$.
3. $A \vee B \to B \vee A$.
4. $(A \to B) \to (C \vee A \to C \vee B)$.

We'll discuss this system in the exercises too.

There are important reasons for studying small formal systems like the axiom systems we've been discussing. Small systems are easier to test and easier to compare with other systems because there are only a few basic operations to worry about. For example, if we build a program to do automatic reasoning, it may be easier to implement a small set of axioms and inference rules. This also applies to computers with small instruction sets and to programming languages with a small number of basic operations.

Exercises

Axiom Systems

1. In the Frege-Lukasiewicz axiom system (6.9), prove the following version of the IP rule: From the premises $\neg A \to B$ and $\neg A \to \neg B$, there is a proof of A. Use only the axioms (6.9), MP, CP, HS, and the six tautologies (6.11).

2. In Frege's axiom system (6.12), prove that Axiom 3 follows from Axioms 1 and 2. That is, prove the following statement from Axioms 1 and 2:

$$(A \to (B \to C)) \to (B \to (A \to C)).$$

3. In Frege's axiom system (6.12), prove that $(\neg A \to \neg B) \to (B \to A)$.

4. In the Hilbert-Ackermann axiom system (6.13), prove each of the following statements. Do not use the CP rule. You can use any previous statement in the list to prove a subsequent statement.

a. $(A \to B) \to ((C \to A) \to (C \to B))$.
b. (HS) If $A \to B$ and $B \to C$ are theorems, then $A \to C$ is a theorem.
c. $A \to A$ (i.e., $\neg A \vee A$).
d. $A \vee \neg A$.
e. $A \to \neg\neg A$.
f. $\neg\neg A \to A$.
g. $\neg A \to (A \to B)$.
h. $(A \to B) \to (\neg B \to \neg A)$.
i. $(\neg B \to \neg A) \to (A \to B)$.

Logic Puzzles

5. The county jail is full. The sheriff, Anne Oakley, brings in a newly caught criminal and decides to make some space for the criminal by letting one of the current inmates go free. She picks prisoners *A, B*, and *C* to choose from. She puts blindfolds on *A* and *B* because *C* is already blind. Next she selects three hats from five hats hanging on the hat rack, two of which are red and three of which are white, and places the three hats on the

prisoner's heads. She hides the remaining two hats. Then she takes the blindfolds off *A* and *B* and tells them what she has done, including the fact that there were three white hats and two red hats to choose from. Sheriff Oakley then says, "If you can tell me the color of the hat you are wearing, without looking at your own hat, then you can go free." The following things happen:

1. *A* says that he can't tell the color of his hat. So the sheriff has him returned to his cell.

2. Then *B* says that he can't tell the color of his hat. So he is also returned to his cell.

3. Then *C*, the blind prisoner, says that he knows the color of his hat. He tells the sheriff, and she sets him free.

What color was *C*'s hat, and how did *C* do his reasoning?

6. Four men and four women were nominated for two positions on the school board. One man and one woman were elected to the positions. Suppose the men are named *A, B, C*, and *D* and the women are named *E, F, G*, and *H*. Further, suppose that the following four statements are true:

1. If neither *A* nor *E* won a position, then *G* won a position.

2. If neither *A* nor *F* won a position, then *B* won a position.

3. If neither *B* nor *G* won a position, then *C* won a position.

4. If neither *C* nor *F* won a position, then *E* won a position.

Who were the two people elected to the school board?

6.5 Chapter Summary

Propositional calculus is the basic building block of formal logic. Each wff represents a statement that can be checked by truth tables to determine whether it is a tautology, a contradiction, or a contingency. There are basic equivalences (6.1) that allow us to simplify and transform wffs into other wffs. We can use these equivalences with Quine's method to determine whether a wff is a tautology, a contradiction, or a contingency. We can also use the equivalences to transform any wff into a DNF or a CNF. Any truth function has one of these forms.

Propositional calculus also provides us with formal techniques for proving properties of wffs without using truth tables. The rules for constructing formal proofs reflect the natural way that we reason informally about the validity of arguments.

We want formal reasoning systems to be sound—proofs yield theorems that are tautologies—and complete—all tautologies can be proven as theorems. The system of antural deduction presented in Section 6.3 is sound and complete. There are even small axiomatic systems for the propositional calculus that are sound and complete. We presented one in Section 6.4 that has three axioms and one inference rule.

Notes

The logical symbols that we've used in this chapter are not universal. So you should be flexible in your reading of the literature. From a historical point of view, Whitehead and Russell [1910] introduced the symbols $\supset$, $\vee$, $\cdot$, $\sim$, and $\equiv$ to stand for implication, disjunction, conjunction, negation, and equivalence, respectively. A prefix notation for the logical operations was introduced by Lukasiewicz [1929], where the letters *C, A, K, N*, and *E* stand for implication, disjunction, conjunction, negation, and equivalence, respectively. So in terms of our notation we have $Cpq = p \to q$, $Apq = p \vee q$, $Kpq = p \wedge q$, $Nq = \neg q$, and $Epq = p \equiv q$. This notation is called Polish notation, and its advantage is that each expression has a unique meaning without using parentheses and precedence. For example, $(p \to q) \to r$ and $p \to (q \to r)$ are represented by the expressions *CCpqr* and *CpCqr*, respectively. The disadvantage of the notation is that it's harder to read. For example, $CCpqKsNr = (p \to q) \to (s \wedge \neg r)$.

The fact that a wff W is a theorem is often denoted by placing a turnstile in front of it as follows:

$$\vdash W.$$

Turnstiles are also used in discussing proofs that have premises. For example, the notation

$$A_1, A_2, \ldots, A_n \vdash B$$

means that there is a proof of B from the premises $A_1, A_2, \ldots, A_n$.

We should again emphasize that the logic we are studying in this book deals with statements that are either true or false. This is sometimes called the *Law of the Excluded Middle*: Every statement is either true or false. If our logic does not assume the law of the excluded middle, then we can no longer use indirect proof because we can't conclude that a statement is false from the assumption that it is not true. A logic called *intuitionist logic* omits this law and thus forces all proofs to be direct. Intuitionists like to construct things in a direct manner.

Logics that assume the law of the excluded middle are called *two-valued logics*. Some logics take a more general approach and consider statements that may not be true or false. For example, in a *three-valued logic* we might use the numbers 0, 0.5, and 1 to stand for the truth values False, Unknown, and True, respectively. We can build truth tables for this logic by defining $\neg A = 1 - A$, $A \vee B = \max(A, B)$, and $A \wedge B = \min(A, B)$. We still use the equivalence $A \rightarrow B \equiv \neg A \vee B$. So we can discuss three-valued logic.

In a similar manner we can discuss *n-valued logic* for any natural number $n \geq 2$, where statements have one of n truth values that can be represented by n numbers in the range 0 to 1. Some logics, called *fuzzy logics*, assign truth values over an infinite set such as the closed unit interval [0, 1].

There are also logics that examine the behavior of "is necessary" and "is possible." Such logics are called *modal logics*, but the term is also used to refer to other logics such as *deontic logic*, which examines the behavior of "is obligated" and "is permitted," and *temporal logic*, which examines the behavior of statements with respect to time such as "has been," "will be," "has always been," and "will always be." For example, the quote at the beginning of this chapter could be represented and analyzed in a modal logic.

All these logics have applications in computer science but they are beyond our scope and purpose. However, it's nice to know that they all depend on a good knowledge of two-valued logic. In this chapter we've covered the fundamental parts of two-valued logic—the properties and reasoning rules of the propositional calculus. We'll see that these ideas occur in all the logics and applications that we cover in the next three chapters.